The Measurement Problem as a Problem of Reference

Determinacy, Measurement Records, and Physically Non-Selective Theories

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Manuscript

Overview. This manuscript treats the measurement problem as a fixed-domain problem of reference to measurement records. The theorem ladder proves a measurement-reference quotient theorem, an equivalence-collapse theorem for same-scope content spaces, an Everettian non-selection theorem, a fixation-commutation hinge theorem for theory-level confirmation, a diachronic inheritance theorem for memory/anticipation/agency, and a selector-exhaustion theorem for interpretation packages.

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The paper is conditional on an explicit fixed downstream substrate. The appendices make the quotient presentation, the imported structural package, and the audit path explicit; they do not replace the main theorem ladder.

Abstract

The measurement problem is usually framed as a problem about outcome definiteness, state reduction, or probability. This paper treats it instead as a fixed-domain problem of reference to measurement records. Work throughout on a downstream regime in which admissibility is already forced, standing is bivalent on the evaluated fragment, same-scope admissibility classification is unique, the admissible interior is unique up to lawful equivalence, and standing is conserved. Within that regime, define admissible reference assignments for diachronic measurement-record discourse and prove a measurement-reference quotient theorem: any admissibility-preserving presentation of the standing-bearing content of such discourse factors uniquely through a quotient of admissible assignments by standing-equivalence. Differences that do not survive that quotient are semantic skin; differences that do survive it are standing-relevant.

The paper then proves an Everettian non-selection theorem for the bare unitary, entangling, decohering package; an equivalence-collapse theorem showing that any same-scope content quotient that preserves standing-bearing use collapses to the measurement-reference quotient; a confirmation criterion stating that theory-level confirmation is well-defined only when lawful redescription preserves that quotient; a fixation-commutation hinge theorem showing that without admissible-redescription commutation theory-level confirmation is undefined rather than merely weakened; a toy witness of hypothesis-order reversal under divergent admissible assignments; a diachronic inheritance theorem for memory, anticipation, deliberation, and confirmation; and an exhaustion theorem stating that interpretation packages can locate the work of singular-reference fixing at exactly one of four structural loci: dynamic selection, ontic privileging, semantic restriction, or discourse deflation.

The paper does not derive the Born rule or alter the Hilbert-space formalism. Its narrower claim is that, on the target explanatory practice, the measurement problem is first a problem of referential determinacy for measurement records and only derivatively a problem of physical outcome production.

Main claim. Measurement records are evidential anchors, not inert marks. On the fixed downstream substrate, admissible reference assignments for those records determine a unique standing-bearing quotient space. Bare Everettian unitary dynamics, entanglement, decoherence, and branch-local record stability do not by themselves collapse the admissible class to one quotient class for diachronic measurement discourse. If such collapse is not supplied elsewhere, theory-level confirmation, memory, anticipation, and deliberation become assignment-relative rather than theory-relative.

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1 Introduction

The measurement problem is usually introduced as a tension among unitary dynamics, the superposition principle, and the apparent definiteness of observed outcomes. On that framing, the central question is how a linear, non-collapsing state description yields a world in which observers report definite results. Collapse theories answer by changing dynamics, hidden-variable theories by adding ontological structure, Everettian theories by retaining unitary dynamics while reinterpreting branching, and instrumentalist or QBist views by weakening the descriptive burden placed on the theory. The debate is then conducted primarily as a dispute over outcome production.[1]

This manuscript isolates a prior problem. Outcome reports enter scientific practice through measurement records. Those records are not inert marks. They anchor claims about what was observed, what is now remembered, what may be anticipated, and how rival hypotheses are to be compared. If a theory is used to underwrite such discourse, then the theory must stabilize reference to those records strongly enough for their evidential role to be fixed. Before asking how one outcome is produced, one must therefore ask what makes a measurement record the determinate referent of the discourse in which it is used.

That question becomes acute in physically non-selective theories. In Everettian quantum mechanics, for example, unitary dynamics and decoherence can stabilize multiple post-measurement branch-histories, each containing a determinate branch-local observer and a determinate branch-local record, without providing a physical selector that privileges one such history as *the* referent of pre-measurement diachronic discourse. Branch-local determinacy is therefore not yet diachronic referential determinacy. The issue is not whether post-branch observers have determinate experiences. The issue is whether the theory fixes which future record is the referent of pre-measurement claims of the form "I will observe o", "I remember observing o", or "this record confirms H_1 over H_2".

The argument of the paper is conditional but exact. Work throughout on a fixed downstream regime in which admissibility is already forced, standing is bivalent on the evaluated fragment, same-scope admissibility classification is unique, the admissible interior is unique up to lawful equivalence, and standing is conserved. Within that fixed regime, define admissible reference assignments for diachronic measurement discourse and consider the standing-bearing content they induce. The central mathematical claim is then a *measurement-reference quotient theorem*: any admissibility-preserving presentation of the reference-bearing content of measurement-record discourse factors uniquely through a quotient of admissible assignments by standing-equivalence. Differences that do not survive that quotient are semantic skin. Differences that do survive it are standing-relevant. The next hardening step is an *equivalence-collapse theorem*: any same-scope quotient relation that claims to preserve standing-bearing content collapses to that same quotient. In particular, divergence that changes confirmational role is not a harmless repackaging of one content; it is quotient-relevant divergence.

Once that quotient structure is in place, the rest of the theorem ladder tightens around one bridge. First, quotient uniqueness and global equivalence collapse show that same-scope standing-bearing content is fixed by the measurement-reference quotient rather than by a discretionary presentation. Second, fixation commutation is identified as the bridge invariant: if lawful redescription does not preserve the relevant quotient class, then the theory fails to preserve its own evidential target. Third, the bare Everettian package is shown to be physically non-selective and therefore not to supply that

[1] For standard landmarks see Einstein et al. [1], Bell [2], Zurek [4], Schlosshauer [5], and Wallace [10].

bridge by dynamics alone. Fourth, theory-level confirmation is proved to be undefined, not merely weakened, whenever fixation fails. Fifth, memory, anticipation, deliberation, and agency inherit the same dependence because any admissible diachronic claim factors through the same quotient-level lineage. Finally, interpretation packages are exhausted by the structural locus at which singular reference is fixed, if it is fixed at all; no fifth same-scope locus remains.

Two restrictions on scope matter throughout. First, the paper is not a general semantics of quantum language. It concerns a narrower target practice: truth-apt diachronic discourse in which measurement records function as evidential anchors for memory, anticipation, deliberation, and theory-level confirmation. Second, the paper does not re-derive the downstream closure package on which it works. That package is imported explicitly and cited where needed; the present manuscript uses it to sharpen the measurement argument rather than to reproduce the companion closure corpus.

The resulting diagnosis is different from the standard presentation of the measurement problem. The first failure is not yet that the theory does not produce a unique outcome. The first failure is that, in a physically non-selective setting, the theory may leave underfixed the referents of the very records through which outcomes are reported and used as evidence. On that target discourse, the measurement problem is therefore a problem of reference to measurement records before it is a problem of outcome production.

Central thesis. Let T be a physically non-selective theory used to underwrite truth-apt diachronic discourse about measurement, memory, anticipation, deliberation, and theory-level confirmation. Then admissible reference assignments for measurement records must either collapse into a single standing-equivalence class on that discourse or else make confirmation and diachronic practice assignment-relative rather than theory-relative. Bare Everettian dynamics, entanglement, decoherence, and branch-local record stability do not by themselves collapse the admissible class. Accordingly, the measurement problem is a problem of referential determinacy for measurement records, not merely a problem of physical outcome production.

The proof order is strict. Section 2 states the fixed downstream substrate, gives a compressed substrate-necessity bridge, and fixes the exact standing conditional. Section 3 defines admissible reference assignments in gate-governed fixed-domain form, proves the measurement-reference quotient theorem, proves the global equivalence-collapse theorem that blocks the charge that the quotient was chosen rather than forced, and then promotes fixation commutation to the bridge invariant between quotient structure and evidential use. Section 4 proves the Everettian non-selection results as the physical trigger for the bridge problem. Section 5 proves the theory-level confirmation criterion and the resulting stabilization consequences. Section 6 exhibits a worked reversal witness. Section 7 proves the diachronic closure and inheritance theorems for memory, anticipation, deliberation, and confirmation. Section 8 derives selector exhaustion from structural constraints before stating the comparative interpretation taxonomy. Section 9 opens with a strict denial table so that standard objections are tied to exact labeled failures. Section 10 marks the manuscript's precise limits. Appendix A supplies a compact formal appendix for the quotient presentation, Appendix B states the exact companion closure package on which the main body is conditional, and Appendix C records the dependency and objection maps.

2 Fixed downstream substrate, target class, and exact conditional

The present manuscript is not foundational in the sense of re-deriving admissibility, standing, reference, or irreversibility from below. It is downstream. The first task is therefore to state explicitly the fixed-domain structural package on which the measurement argument runs.

2.1 Fixed downstream substrate

Definition 2.1 (Fixed downstream substrate)**.** A *fixed downstream substrate* for the present manuscript is a same-scope, admissibility-governed regime in which the following are already in force.

1. Admissibility is a forced governance boundary for the target class of non-degenerate, proposition-bearing reasoning.
2. Standing is bivalent on the evaluated fragment and fail-closed when evaluation is not instantiated.
3. Same-scope admissibility classification is unique: there is no second admissibility predicate, no admissibility-relevant boundary selector, and no admissibility-relevant parameterization on the fixed scope.
4. The admissible interior is unique up to lawful equivalence.
5. Standing is conserved: there is no same-scope repair, no generator, no primitive carrier, and no same-scope transfer without fresh target gating.
6. Any second same-scope invariant either collapses to bookkeeping or constitutes a scope change.

Remark 2.2 (Provenance of the substrate)**.** Definition 2.1 is not a hidden premise. It is the explicit fixed-domain package proved in the companion closure papers on kernel necessity, interface forcing, trajectory closure, uniqueness, and structural singleton results; see [16–23]. The present paper uses that package as background and does not attempt to re-prove it in full. What it does add locally is a compressed bridge theorem showing why, once the target explanatory burden is fixed, this substrate is not optional for the measurement argument.

2.2 Compressed substrate-necessity bridge

Definition 2.3 (Minimal coherence triggers)**.** A same-scope discourse-practice satisfies the *minimal coherence triggers* relevant to the present paper iff it preserves reference under lawful redescription, treats admissible continuation as compositionally available candidate material, carries non-trivial inferential content, and executes at least one irreversible commitment on the fixed scope.

Lemma 2.4 (Stable reference is necessary for the target explanatory practice)**.** *If measurement records are used as theory-level evidential anchors under lawful same-scope redescription, then lawful redescription must preserve what those records are about. Otherwise the same evidential claim changes target under admissible use and no standing-bearing evidential role is fixed.*

Proof. A theory-level evidential claim is supposed to remain the same claim across lawful same-scope redescription. If lawful redescription changes what the claim is about, then the evidential role of the

record is presentation-dependent rather than theory-relative. On the fail-closed substrate recorded in Definition 2.1, that is not a weaker kind of confirmation; it is loss of determinate standing-bearing use. So stable reference is necessary. □

Lemma 2.5 (Identity quotient necessity)**.** *If same-scope measurement-record discourse is stable under lawful redescription, then admissible reference assignments admit a standing-equivalence relation whose classes exhaust same-scope standing-bearing content for the target discourse. Any same-scope standing-bearing presentation of that content therefore factors through a quotient of admissible assignments.*

Proof. Once lawful redescription preserves standing-bearing use, assignments that agree on every standing-bearing value in the target discourse cannot be treated as distinct same-scope contents unless one introduces an additional same-scope invariant. But the fixed downstream substrate excludes any second same-scope invariant except as bookkeeping or scope change. Therefore standing-bearing content must collapse to equivalence classes of assignments, and any same-scope standing-bearing presentation must factor through the corresponding quotient. □

Lemma 2.6 (Irreversibility forces bivalent gating)**.** *If same-scope illicit evaluated acts cannot be repaired as the same acts, then evaluated acts must fall under a fail-closed eligible/ineligible gate on the evaluated fragment. No third standing-relevant status remains.*

Proof. Without a gate there is no stable distinction between eligible and ineligible evaluated acts, so irreversibility has nothing determinate to attach to. If a third standing-relevant status were admitted, then some acts would be neither fully eligible nor fully ineligible and illegitimacy could be softened or reclassified without a fresh gate. That is exactly the kind of same-act repair excluded by the irreversibility condition. Hence irreversibility forces bivalent fail-closed gating. □

Lemma 2.7 (Bivalence plus the AMetric boundary excludes parameterization)**.** *If the admissibility interface is bivalent on the evaluated fragment and the pre-admissible side is AMetric, then no same-scope admissibility-relevant parameter, ranking, or selector can refine the boundary.*

Proof. A same-scope admissibility-relevant parameter would introduce more than the two bivalent statuses of eligible and ineligible, or else it would provide a selector among allegedly admissible alternatives. Either way the boundary would cease to be AMetric and non-parameterized. So bivalence together with the AMetric boundary excludes same-scope admissibility-relevant parameterization. □

Theorem 2.8 (Compressed substrate-necessity bridge)**.** *Let a same-scope discourse-practice satisfy the minimal coherence triggers of Definition 2.3 and let it be used to underwrite the target explanatory practice of Definition 2.12. Then the fixed downstream substrate of Definition 2.1 is not optional: admissibility is forced, standing is bivalent on the evaluated fragment, same-scope admissibility classification is unique, the admissible interior is unique up to lawful equivalence, standing is conserved, and any second same-scope invariant collapses to bookkeeping or scope change.*

Proof. Lemma 2.4 forces stable reference for the target explanatory practice. Lemma 2.5 then forces quotient-level standing-bearing content rather than representative-dependent content. Lemma 2.6 forces fail-closed bivalent gating on the evaluated fragment, and Lemma 2.7 excludes same-scope

admissibility- relevant boundary parameterization or selection. The remaining fixed-domain consequences — unique same-scope admissibility classification, unique admissible interior, standing conservation, and the collapse of any second same-scope invariant — are exactly the downstream closure results proved in [16–23]. Therefore, once the target explanatory practice is fixed on a same scope, the substrate recorded in Definition 2.1 is forced rather than optional. □

Proposition 2.9 (Immediate consequences of the substrate)**.** *Within the fixed downstream substrate, any same-scope admissibility-bearing distinction about measurement-record discourse must appear in one of only two places:*

1. *in the standing classification of evaluated acts, or*
2. *in the licensed continuation structure of the discourse built on those acts.*

No third same-scope locus of admissibility-bearing work remains.

Proof. By Definition 2.1, same-scope admissibility classification is unique, standing is conserved, and no second same-scope invariant survives except as bookkeeping or scope change. Therefore any admissibility-bearing difference must either change which acts stand, or change what same-scope continuation is licensed for standing-bearing acts. If it changes neither, then by the same fixed-domain package it is semantic or representational surplus rather than admissibility-bearing content. So no third locus remains. □

2.3 Physically non-selective theories and the referential selection problem

Definition 2.10 (Physical non-selection)**.** A theory is *physically non-selective* with respect to a measurement process if, after the measurement interaction, the theory's physical description realizes multiple successor histories containing stable measurement records and physically continuous observer-successors, while supplying no physical law, state variable, or dynamically privileged trajectory that singles out one such successor history as physically distinguished from the others.

Definition 2.11 (Diachronic referential selection problem)**.** Given a physically non-selective theory, the *diachronic referential selection problem* is the problem whether the theory, together with its admissible interpretive use on the fixed downstream substrate, determines which successor history is the referent of pre-measurement diachronic discourse.

Definitions 2.10 and 2.11 separate two questions that are often run together. Physical non-selection is a claim about what the physical theory itself delivers: multiplicity without a physically distinguishing selector. The diachronic referential selection problem is the further question whether that physical package, together with whatever admissible same-scope use it supports, determines which successor history is the referent of discourse about what the pre-measurement subject will observe, remember, or use as evidence.

Everettian quantum mechanics is the paradigm case. Unitary dynamics and decoherence can stabilize multiple post-measurement branch-histories, each carrying a determinate branch-relative observer and a determinate branch-relative record, while leaving the physical theory itself non-selective among them. The issue is not whether the theory assigns numerical weights to those histories. The issue is whether the theory itself fixes which physically realized history is the referent of the measurement record discourse of an internal subject.

2.4 Exact conditional and target discourse

The manuscript runs under one standing conditional.

Standing conditional. If a quantum theory is used to underwrite truth-apt diachronic discourse about measurement, memory, anticipation, deliberation, and evidential confirmation, then admissible reference assignments for measurement records must either

1. collapse into a single standing-bearing class on that discourse, or
2. leave confirmation and diachronic practice assignment-relative rather than theory-relative.

This conditional is not a disguised collapse postulate. It says only that if a theory is assigned explanatory work involving records as evidence, then the referents of those records cannot vary in same-scope standing-bearing ways while the theory still claims a single theory-level evidential output.

Definition 2.12 (Target explanatory practice)**.** The *target explanatory practice* of this paper is the practice in which measurement records function as evidential anchors for theory-level claims about what was observed, what follows for memory and anticipation, and how those observations bear on rival hypotheses.

Definition 2.13 (Fixed scope of the paper)**.** The fixed scope of the paper is the same-scope, discourse-committing use of a physically non-selective measurement theory within the target explanatory practice of Definition 2.12. Moves that import new ontology, new dynamics, new admissibility-relevant structure, new boundary selectors, or new target gating count as scope changes rather than same-scope repairs.

2.5 Public vocabulary

The main body stays close to standard foundations vocabulary while keeping the fixed-domain background explicit.

- **Measurement record**: a physically realized outcome-bearing structure used as an evidential anchor.
- **Diachronic measurement discourse**: discourse linking a pre-measurement subject to post-measurement records by way of memory, anticipation, deliberation, or confirmation.
- **Reference assignment**: a map from diachronic measurement claims to candidate branch-record referents.
- **Admissible reference assignment**: a same-scope reference assignment eligible under the gate-governed fixed-domain constraints stated in Section 3.
- **Selector**: any principle that fixes singular reference by privileging one continuation, history, or record as the referent relevant to the target discourse.
- **Standing-bearing content**: content that survives into standing classification or licensed continuation on the fixed downstream substrate.

- **Skin**: representational difference that does not survive into standing-bearing content.

2.6 Scope of the present manuscript

Theorems in the present manuscript do not settle every metaphysical question in quantum foundations. Their function is narrower. On the fixed downstream substrate,

1. the paper identifies the measurement problem as a fixed-domain problem of reference to measurement records,
2. proves a quotient uniqueness theorem for admissible measurement-reference content,
3. shows that the bare Everettian package is physically non-selective with respect to that target discourse,
4. proves that confirmational divergence among admissible assignments is quotient-relevant and therefore not mere semantic skin,
5. proves that theory-level confirmation requires fixation commutation under admissible redescription,
6. proves that memory, anticipation, and agency inherit the same lineage dependence, and
7. classifies interpretation packages by the structural locus at which singular reference is fixed, if it is fixed at all.

The manuscript is therefore neither a general semantics of quantum theory nor a re-derivation of the companion closure corpus. It is a fixed-domain measurement-reference paper built on that corpus.

3 Measurement-reference quotient and minimal formal setting

This section fixes the minimum formal vocabulary required by the proof bridge. The key upgrade is a quotient theorem. Admissible reference assignments are treated as a same-scope parameter space. Standing-bearing differences among them are then separated from semantic skin by an explicit quotient construction.

We work with a fixed measurement setting consisting of a pre-measurement subject-stage s_0, a nonempty set $\mathcal{B}$ of physically realized post-measurement histories, a record map $\mathsf{Rec} : \mathcal{B} \to \mathcal{R}$, a continuant map $\mathsf{Cont} : \mathcal{B} \to \mathcal{C}$, a set $\mathcal{D}$ of diachronic measurement claims, a set $\mathcal{H}$ of rival hypotheses, a set $\mathcal{E}$ of evidential propositions, and a confirmational-role assignment

$$\mathsf{CR} : \mathcal{E} \to (\mathcal{H} \times \mathcal{H} \to \{-1, 0, 1\}).$$

Here 1 means that the evidential proposition favors the first hypothesis over the second, -1 means that it favors the second over the first, and 0 means that it leaves their comparative ordering unchanged. This is only a minimal comparative scaffold; the paper does not presuppose a full confirmation theory.

Definition 3.1 (Measurement record)**.** A *measurement record* is a physically realized structure that the target explanatory practice uses as an evidential anchor for claims about what outcome was observed.

A *diachronic measurement claim* is any claim whose truth conditions depend on a relation between the pre-measurement subject-stage s_0 and one or more post-measurement record-bearing histories. Typical examples are claims of the form

"I will observe outcome o," "I remember observing outcome o," or "this record confirms H_i over H_j."

For a claim $d \in \mathcal{D}$, let $\mathsf{Cand}(d) \subseteq \mathcal{B} \times \mathcal{R}$ be the set of candidate branch-record pairs physically realized by the theory and eligible to serve as referents for d.

Definition 3.2 (Reference assignment)**.** A *reference assignment* is a rule

$$\rho : \mathcal{D} \to \mathcal{B} \times \mathcal{R}$$

that maps each diachronic measurement claim $d \in \mathcal{D}$ to a candidate branch-record pair $\rho(d) \in \mathsf{Cand}(d)$.

Definition 3.3 (Admissible reference assignment)**.** A reference assignment ρ is *admissible* on the fixed scope of Definition 2.13 iff it satisfies all of the following.

1. **Same-scope condition.** ρ interprets the claims in $\mathcal{D}$ on the fixed downstream substrate and does not rely on scope reset, fresh target gating, or post hoc domain change.
2. **Ontology-compatibility.** For every $d \in \mathcal{D}$, $\rho(d)$ selects only a physically realized candidate in $\mathsf{Cand}(d)$.
3. **Discourse-compatibility.** ρ is suitable to interpret the target explanatory practice, namely truth-apt discourse involving measurement, memory, anticipation, deliberation, and evidential confirmation.
4. **Non-question-begging use.** Admissibility of ρ is not defined by stipulating in advance the very confirmational conclusion the later theorems are meant to establish.
5. **No second classifier / no boundary selector.** The admissibility of ρ does not depend on introducing a second same-scope admissibility predicate, a boundary selector, or an admissibility-relevant parameterization external to the fixed downstream substrate.

The set of all admissible reference assignments for the fixed setting is written ARef.

Clause (5) is the crucial strengthening. It prevents the assignment space from being widened by a same-scope selector principle or narrowed by a second admissibility classifier whose only role would be to manufacture uniqueness.

3.1 Standing-bearing profiles of admissible assignments

For $\rho \in \mathsf{ARef}$ and $d \in \mathcal{D}$, let $\mathsf{Prop}_\rho(d) \in \mathcal{E}$ denote the evidential proposition induced by taking $\rho(d)$ as the referent of d whenever d is used evidentially. More generally, let

$$\mathsf{Val}_\rho(d)$$

be the standing-bearing discourse value of d under ρ: for ordinary truth-apt diachronic claims it is the induced truth value, and for evidential claims it is the induced comparative evidential role. The target-discourse profile of ρ is

$$\mathsf{Prof}(\rho) := \big(\mathsf{Val}_\rho(d)\big)_{d\in\mathcal{D}}.$$

Definition 3.4 (Confirmational-role equivalence)**.** Two admissible reference assignments $\rho, \sigma \in \mathsf{ARef}$ are *confirmationally equivalent*, written $\rho \approx_{\text{role}} \sigma$, if for every evidential claim $d \in \mathcal{D}$,

$$\mathsf{CR}(\mathsf{Prop}_\rho(d)) = \mathsf{CR}(\mathsf{Prop}_\sigma(d)).$$

Equivalently, ρ and σ are confirmationally equivalent exactly when every theory-level evidential claim licensed under one assignment has the same comparative confirmational role under the other.

Lemma 3.5 (Confirmational-role equivalence is an equivalence relation)**.** *The relation* $\approx_{\text{role}}$ *on* ARef *is reflexive, symmetric, and transitive.*

Proof. Reflexivity holds because for every admissible assignment ρ and every evidential claim d, we have $\mathsf{CR}(\mathsf{Prop}_\rho(d)) = \mathsf{CR}(\mathsf{Prop}_\rho(d))$. Symmetry is immediate because equality of confirmational roles is symmetric. Transitivity holds because if $\mathsf{CR}(\mathsf{Prop}_\rho(d)) = \mathsf{CR}(\mathsf{Prop}_\sigma(d))$ and $\mathsf{CR}(\mathsf{Prop}_\sigma(d)) = \mathsf{CR}(\mathsf{Prop}_\tau(d))$ for every relevant d, then $\mathsf{CR}(\mathsf{Prop}_\rho(d)) = \mathsf{CR}(\mathsf{Prop}_\tau(d))$ for every such d. □

Definition 3.6 (Measurement-reference standing equivalence)**.** Two admissible reference assignments $\rho, \sigma \in \mathsf{ARef}$ are *standing-equivalent for the target discourse*, written

$$\rho \approx_{\text{MR}} \sigma,$$

iff

$$\mathsf{Prof}(\rho) = \mathsf{Prof}(\sigma).$$

Thus $\rho \approx_{\text{MR}} \sigma$ exactly when the two assignments agree on every standing-bearing value of every claim in the target explanatory practice.

Definition 3.7 (Skin-equivalent versus standing-relevant divergence)**.** A difference between admissible reference assignments is *semantic skin* iff the assignments lie in the same $\approx_{\text{MR}}$-class. It is *standing-relevant divergence* iff they lie in distinct $\approx_{\text{MR}}$-classes.

3.2 Presentations of measurement-reference content

Definition 3.8 (Measurement-reference presentation)**.** A *measurement-reference presentation* is a triple (X, r, Val_X) where:

1. X is a set intended to serve as the standing-bearing content space for admissible reference assignments on the fixed scope;
2. $r : \mathsf{ARef} \to X$ is a surjection; and
3. for every claim $d \in \mathcal{D}$ there exists a map

$$\mathsf{Val}_X^d : X \to \mathcal{V}_d$$

for some codomain $\mathcal{V}_d$ such that

$$\mathsf{Val}_\rho(d) = \mathsf{Val}_X^d(r(\rho)) \qquad (\rho \in \mathsf{ARef}).$$

The presentation is *sound* if $\rho \approx_{\mathrm{MR}} \sigma$ implies $r(\rho) = r(\sigma)$, and *complete* if $r(\rho) = r(\sigma)$ implies $\rho \approx_{\mathrm{MR}} \sigma$.

Lemma 3.9 (Completeness of measurement-reference presentations)**.** *Every measurement-reference presentation is complete.*

Proof. Assume $r(\rho) = r(\sigma)$. Then for every $d \in \mathcal{D}$,

$$\mathsf{Val}_\rho(d) = \mathsf{Val}_X^d(r(\rho)) = \mathsf{Val}_X^d(r(\sigma)) = \mathsf{Val}_\sigma(d).$$

Hence $\mathsf{Prof}(\rho) = \mathsf{Prof}(\sigma)$ and therefore $\rho \approx_{\mathrm{MR}} \sigma$. □

Lemma 3.10 (Soundness forced by stable reference and fixed-domain uniqueness)**.** *Within the fixed downstream substrate, every measurement-reference presentation is sound.*

Proof. Assume $\rho \approx_{\mathrm{MR}} \sigma$ but $r(\rho) \neq r(\sigma)$. Then the presentation distinguishes two putative same-scope standing-bearing contents that agree on every value in the target explanatory practice. By Definition 2.1 and Proposition 2.9, any same-scope admissibility-bearing distinction must survive either into standing classification or into licensed continuation. But $\rho \approx_{\mathrm{MR}} \sigma$ means that the assignments agree on the full target-discourse profile: there is no difference in standing-bearing use for memory, anticipation, deliberation, or confirmation. Treating $r(\rho)$ and $r(\sigma)$ as distinct therefore introduces an unwitnessed same-scope distinction. Within the fixed downstream substrate, such a distinction is either bookkeeping only or a scope change. Since X is presented as a standing-bearing same-scope content space, only bookkeeping remains. Hence $r(\rho) = r(\sigma)$ is forced. □

Lemma 3.11 (Quotient universal property)**.** *Let $q : \mathsf{ARef} \to \mathsf{ARef}/\approx_{\mathrm{MR}}$ be the quotient map. If $F : \mathsf{ARef} \to Y$ is constant on $\approx_{\mathrm{MR}}$-classes, then there exists a unique map $\widetilde{F} : \mathsf{ARef}/\approx_{\mathrm{MR}} \to Y$ such that*

$$F = \widetilde{F} \circ q.$$

Proof. Define $\widetilde{F}([\rho]) := F(\rho)$. This is well-defined because F is constant on $\approx_{\mathrm{MR}}$-classes. Uniqueness follows because q is surjective. □

Theorem 3.12 (Measurement-reference quotient theorem)**.** *Let (X, r, Val_X) be any measurement-reference presentation on the fixed downstream substrate. Then r is automatically sound and complete and induces a unique bijection*

$$\phi : \mathsf{ARef}/\approx_{\mathrm{MR}} \longrightarrow X, \qquad \phi([\rho]) = r(\rho).$$

In particular, the quotient $\mathsf{ARef}/\approx_{\mathrm{MR}}$ is the unique standing-bearing content space for admissible measurement-reference assignments on the fixed scope, up to unique isomorphism.

Proof. Completeness is Lemma 3.9. Soundness is Lemma 3.10. Define $\phi([\rho]) := r(\rho)$. Soundness implies that ϕ is well-defined. Completeness implies that ϕ is injective. Surjectivity holds because r is surjective. Uniqueness is exactly the quotient universal property of Lemma 3.11: r is constant on $\approx_{\mathrm{MR}}$-classes by soundness and therefore factors through q uniquely. □

Definition 3.13 (Admissibility-preserving equivalence relation)**.** An equivalence relation $\approx$ on ARef is *admissibility-preserving on the fixed scope* iff both of the following hold:

1. every standing-bearing value on the fixed scope factors through the quotient map

$$q_{\approx} : \mathsf{ARef} \to \mathsf{ARef}/{\approx};$$

   equivalently, $(\mathsf{ARef}/{\approx}, q_{\approx}, \mathsf{Val}_{\approx})$ is a same-scope measurement-reference presentation in the sense of Definition 3.8; and

2. the quotient is still claimed to carry same-scope standing-bearing content rather than mere bookkeeping.

Theorem 3.14 (Global equivalence-collapse theorem)**.** *Let $\approx$ be any admissibility-preserving equivalence relation on ARef. Then for all $\rho, \sigma \in \mathsf{ARef}$,*

$$\rho \approx \sigma \iff \rho \approx_{\mathrm{MR}} \sigma.$$

Equivalently, any admissibility-preserving same-scope equivalence relation on measurement-reference assignments coincides with standing-equivalence. The quotient $\mathsf{ARef}/\approx_{\mathrm{MR}}$ is therefore forced, not chosen.

Proof. Assume first that $\rho \approx \sigma$ but $\rho \not\approx_{\mathrm{MR}} \sigma$. Then $\mathsf{Prof}(\rho) \neq \mathsf{Prof}(\sigma)$, so there exists a claim $d \in \mathcal{D}$ with $\mathsf{Val}_\rho(d) \neq \mathsf{Val}_\sigma(d)$. But by Definition 3.13, every standing-bearing value on the fixed scope factors through $q_{\approx}$. Since $q_{\approx}(\rho) = q_{\approx}(\sigma)$, the induced value of d would have to agree on ρ and σ, contradiction. So $\approx$ cannot be strictly coarser than $\approx_{\mathrm{MR}}$.

Assume next that $\rho \approx_{\mathrm{MR}} \sigma$ but $\rho \not\approx \sigma$. Then $\approx$ distinguishes two assignments that agree on every standing-bearing value in the target discourse. Treating that additional distinction as same-scope content would amount to a second closure-relevant equivalence relation finer than standing-equivalence. By the fixed downstream substrate and Theorem 3.12, no such second same-scope equivalence survives: it is either bookkeeping only or a scope change. Since $\approx$ is assumed admissibility-preserving, that option is unavailable. So $\approx$ cannot be strictly finer than $\approx_{\mathrm{MR}}$.

Therefore neither a coarser nor a finer admissibility-preserving same-scope equivalence relation is possible, and the two relations coincide. □

Corollary 3.15 (No coarser admissibility-preserving equivalence)**.** *No admissibility-preserving same-scope equivalence relation on ARef can identify assignments that lie in distinct $\approx_{\mathrm{MR}}$-classes.*

Proof. That would make the equivalence strictly coarser than $\approx_{\mathrm{MR}}$, contradicting Theorem 3.14. □

Corollary 3.16 (No finer admissibility-preserving equivalence)**.** *No admissibility-preserving same-scope equivalence relation on ARef can distinguish assignments that lie in the same $\approx_{\mathrm{MR}}$-class.*

Proof. That would make the equivalence strictly finer than $\approx_{\mathrm{MR}}$, contradicting Theorem 3.14. □

Corollary 3.17 (Alternative same-scope equivalence relations collapse)**.** *No same-scope equivalence relation on ARef can preserve standing-bearing measurement-reference content while remaining genuinely distinct from $\approx_{\mathrm{MR}}$. Any such purported alternative either collapses to $\approx_{\mathrm{MR}}$ or else ceases to be admissibility-preserving on the fixed scope.*

Proof. Immediate from Theorem 3.14 together with Corollaries 3.15 and 3.16. □

Corollary 3.18 (Confirmational divergence is quotient-relevant)**.** *If there exist $\rho, \sigma \in \mathsf{ARef}$ and an evidential claim $d \in \mathcal{D}$ such that*

$$\mathsf{CR}(\mathsf{Prop}_\rho(d)) \neq \mathsf{CR}(\mathsf{Prop}_\sigma(d)),$$

then $[\rho] \neq [\sigma]$ in $\mathsf{ARef}/\approx_{\mathrm{MR}}$. In particular, confirmational divergence is not semantic skin.

Proof. If $[\rho] = [\sigma]$, then by definition $\rho \approx_{\mathrm{MR}} \sigma$, so $\mathsf{Prof}(\rho) = \mathsf{Prof}(\sigma)$. That in particular forces equality of all confirmational-role values on evidential claims. The displayed inequality therefore implies $[\rho] \neq [\sigma]$. □

3.3 Fixation as the bridge invariant

Definition 3.19 (Lawful redescription of an evidential claim)**.** A *lawful redescription* of an evidential claim $d \in \mathcal{D}$ is a same-scope re-expression of d that preserves its candidate set $\mathsf{Cand}(d)$, its target discourse role, and the fixed downstream substrate. In particular, lawful redescription introduces no new selector, no new admissibility classifier, and no scope reset.

Definition 3.20 (Fixation commutation)**.** Fixation *commutes with admissible redescription* for an evidential claim d iff every lawful redescription of d preserves its measurement-reference quotient class. Equivalently, lawful redescription cannot move the evidential use of d between distinct standing-bearing admissible assignment classes.

Lemma 3.21 (Failed fixation commutation is quotient movement)**.** *If fixation fails to commute with admissible redescription for an evidential claim d, then there exist lawful redescriptions of d whose admissible uses lie in distinct $\approx_{\mathrm{MR}}$-classes.*

Proof. This is just the negation of Definition 3.20: failure of commutation means that lawful redescription moves the evidential use of d between distinct standing-bearing admissible assignment classes. □

Theorem 3.22 (Fixation-commutation necessity for theory-level confirmation)**.** *If fixation fails to commute with admissible redescription for an evidential claim d, then theory-level confirmation through d is undefined on the fixed scope.*

Proof. By Lemma 3.21, failure of fixation commutation moves the admissible use of d between distinct standing-bearing quotient classes. By Theorem 3.14, that movement cannot be absorbed by an alternative same-scope equivalence relation; it is genuine standing-bearing divergence. A theory-level confirmational verdict, however, is supposed to be attributable to the theory on the fixed scope rather than to a further standing-bearing choice among admissible quotient classes. So once lawful redescription moves d between distinct quotient classes, no theory-relative confirmational verdict is fixed. The right description is therefore undefinedness of theory-level confirmation on the fixed scope, not merely a weaker or relativized verdict. □

Corollary 3.23 (Fixation commutation is the bridge invariant)**.** *On the fixed scope, quotient uniqueness and equivalence collapse become relevant to measurement only through fixation commutation: if*

lawful redescription preserves the measurement-reference quotient class of an evidential claim, then same-scope standing-bearing divergence is silent for that use; if it does not, theory-level confirmation is undefined by Theorem 3.22.

Proof. The quotient theorem fixes the unique same-scope content space, and Theorem 3.14 closes alternative same-scope equivalences. The only remaining question for theory-level evidential use is whether lawful redescription preserves the relevant quotient class. When it does, same-scope standing-bearing divergence is silent for that use; when it does not, Theorem 3.22 applies. □

Remark 3.24 (Status of admissibility constraints)**.** The admissibility condition is intentionally modest but not empty. Clause (5) of Definition 3.3 closes the easiest hostile move: enlarging or pruning the assignment space by smuggling in a second same-scope admissibility classifier or a boundary selector. Theorem 3.12 then shows that once the assignment space is typed that way, only quotient-level differences count as standing-bearing, and Theorem 3.14 shows that this quotient is not a discretionary choice among many same-scope quotients but the unique admissibility-preserving one. Theorem 3.22 then makes fixation commutation the bridge from quotient structure to the evidential use of measurement records.

4 The Everettian diagnosis

This section introduces a minimal Everettian measurement package and asks whether bare unitary branching supplies the fixation bridge isolated in Section 3. It proves that Everettian physics is non-selective in the sense fixed in Section 2, separates branch-local determinacy from diachronic reference, and shows that decoherence, centered descriptions, self-location, and decision theory do not by themselves fix the referent of pre-measurement discourse.

4.1 The Everettian base package

The paper needs only a small amount of quantum mechanics. For the present purpose an Everettian measurement episode can be modeled by a pre-measurement state, an entangling measurement interaction, and a decoherence regime that renders the resulting outcome-sectors approximately dynamically autonomous for the duration of the measurement record.

Definition 4.1 (Everettian base package for a measurement episode)**.** A measurement episode satisfies the *Everettian base package* when all of the following hold:

1. **Unitary evolution.** The total post-interaction state is obtained by a unitary measurement interaction U_M applied to the pre-measurement state.

2. **Outcome-sector multiplicity.** There exists a decomposition

$$\Psi_M = \sum_{i \in I} \alpha_i \psi_i, \qquad |I| \geq 2, \qquad \alpha_i \neq 0 \text{ for each } i,$$

   where each ψ_i is associated with a distinct stable record/outcome package $(\mathsf{Cont}_i, \mathsf{Rec}_i, o_i)$.

3. **Decoherence.** The sectors ψ_i are pairwise approximately non-interfering for the purposes of the measurement record, so that each supports branch-local record stability.

4. **No selector in the bare physics.** No additional collapse postulate, hidden-variable actualization rule, or other branch-privileging physical structure is added to the bare unitary description.

The definition deliberately builds in only what an Everettian appeals to when claiming that unitary quantum theory together with decoherence already explains measurement: global unitary evolution, multiple nonzero outcome-sectors, and record stability without a collapse or actualization rule [3, 4, 10].

Lemma 4.2 (Branch-local determinacy)**.** *Suppose a measurement episode satisfies Definition 4.1. Then each decohered outcome-sector contains a determinate branch-local record and a determinate branch-local observer-state.*

Proof. By clause (2) of Definition 4.1, each sector ψ_i is associated with a stable record/outcome package $(\mathsf{Cont}_i, \mathsf{Rec}_i, o_i)$. By clause (3), decoherence suppresses interference between the sectors for the purposes of the measurement record, so each such package is dynamically stable in the branch-local sense relevant to reporting, memory, and within-branch experience. Hence each decohered sector contains a determinate branch-local record and branch-local observer-state. □

Lemma 4.3 (Unitary multiplicity lemma)**.** *Suppose a measurement episode satisfies Definition 4.1. Then the post-measurement state contains at least two nonzero physically realized outcome-sectors.*

Proof. By clause (2) of Definition 4.1, the post-measurement state has the form

$$\Psi_M = \sum_{i \in I} \alpha_i \psi_i$$

with $|I| \geq 2$ and $\alpha_i \neq 0$ for each $i \in I$. A nonzero coefficient means that the corresponding sector contributes nontrivially to the physical state-description of the episode. Since there are at least two such coefficients, the post-measurement state contains at least two nonzero outcome-sectors. By the intended Everettian reading, those sectors are physically realized parts of the same global state rather than merely hypothetical alternatives. Hence the episode contains at least two nonzero physically realized outcome-sectors. □

Lemma 4.4 (Entanglement correlates without privileging)**.** *Let a measurement episode satisfy Definition 4.1. Then the entangling measurement interaction correlates system, apparatus, and observer records across the distinct outcome-sectors, but does not by itself privilege one such sector as the unique diachronic continuation relevant to pre-measurement measurement discourse.*

Proof. In the Everettian package, the measurement interaction produces a correlated state of the schematic form

$$\sum_{i \in I} \alpha_i \, |s_i\rangle |A_i\rangle |O_i\rangle,$$

where distinct system-outcome states are correlated with distinct apparatus and observer-record states. That establishes the correlation claim: each sector carries its own determinate branch-local triple and hence its own record/outcome package.

What must be shown is the absence of privileging. The entangling state specifies, for each i, what the record is in that sector. It does not specify that one i is the unique referent of pre-measurement

first-person discourse. To privilege one sector, the physical theory would need an additional rule of the form: exactly one correlated sector is the continuation relevant to the pre-measurement subject. But clause (4) of Definition 4.1 excludes any such added selector in the bare physics. Therefore entanglement supplies correlated multiplicity, not a unique diachronic referent. □

Lemma 4.5 (Decoherence yields stability, not singularity)**.** *If a measurement episode satisfies Definition 4.1, then decoherence can explain the stability and effective autonomy of each branch-local record without thereby selecting which branch is the referent of pre-measurement diachronic discourse.*

Proof. By clause (3) of Definition 4.1, decoherence renders the sectors approximately non-interfering for the purposes of the measurement record. This yields the standard Everettian achievement: each sector supports a stable branch-local record and observer-state.

However, branch-local stability is a within-sector property. It tells us that if a sector is taken as the relevant one, its record is robust enough for local memory and reporting. It does not add any physical fact of the form "sector i rather than sector j is the unique continuation relevant to pre-measurement discourse". Such a fact would be a selector. Since clause (4) excludes selectors in the bare physics, decoherence yields stability but not singularity of referent. □

Lemma 4.6 (No supervenient diachronic selector)**.** *Let T be physically non-selective with respect to a measurement process, and let O_i and O_j be two post-measurement observer-successors realized by T. Suppose that T supplies no physical law, state variable, dynamically privileged trajectory, or physically relevant asymmetry that distinguishes O_i from O_j with respect to diachronic continuation. Then no unique diachronic continuation relation privileging O_i over O_j supervenes on the physical facts delivered by T alone.*

Proof. To say that a unique diachronic continuation relation supervenes on the physical facts delivered by T is to say that those facts suffice to determine, without further addition, that one successor is the privileged continuation rather than the other. But by hypothesis, the physical description delivered by T contains no law, state variable, privileged trajectory, or other physically relevant asymmetry distinguishing O_i from O_j with respect to diachronic continuation. Any relation that nevertheless privileges O_i over O_j would therefore introduce an asymmetry not fixed by the physical facts supplied by T. Hence no such unique privileging relation supervenes on the physical facts delivered by T alone. □

Corollary 4.7 (Everettian non-selection)**.** *In a branching Everettian measurement model satisfying physical non-selection, the physical structure alone determines neither an explicit selector nor a supervenient unique diachronic referent for pre-measurement first-person discourse.*

Proof. By Lemma 4.3, the Everettian base package contains at least two nonzero physically realized outcome-sectors, each with its own branch-local record and continuant. By Definition 2.10, the package contains no physical law, state variable, or dynamically privileged trajectory distinguishing one such successor as physically privileged. The absence of an explicit selector is therefore immediate. Applying Lemma 4.6 to any two such realized successors yields the absence of a supervenient unique diachronic referent. □

Theorem 4.8 (Everettian non-selection theorem)**.** *Every measurement episode satisfying the Everettian base package is physically non-selective in the sense of Definition 2.10.*

Proof. To be physically non-selective, a measurement episode must realize multiple stable successor histories while supplying no physical law, state variable, or dynamically privileged trajectory that singles out one such history as physically distinguished. The first condition follows from Lemma 4.3; branch-local determinacy of the realized histories follows from Lemma 4.2; and the absence of a physically distinguishing selector is built into clause (4) of Definition 4.1 and then made explicit by Corollary 4.7. Hence every measurement episode satisfying the Everettian base package is physically non-selective. □

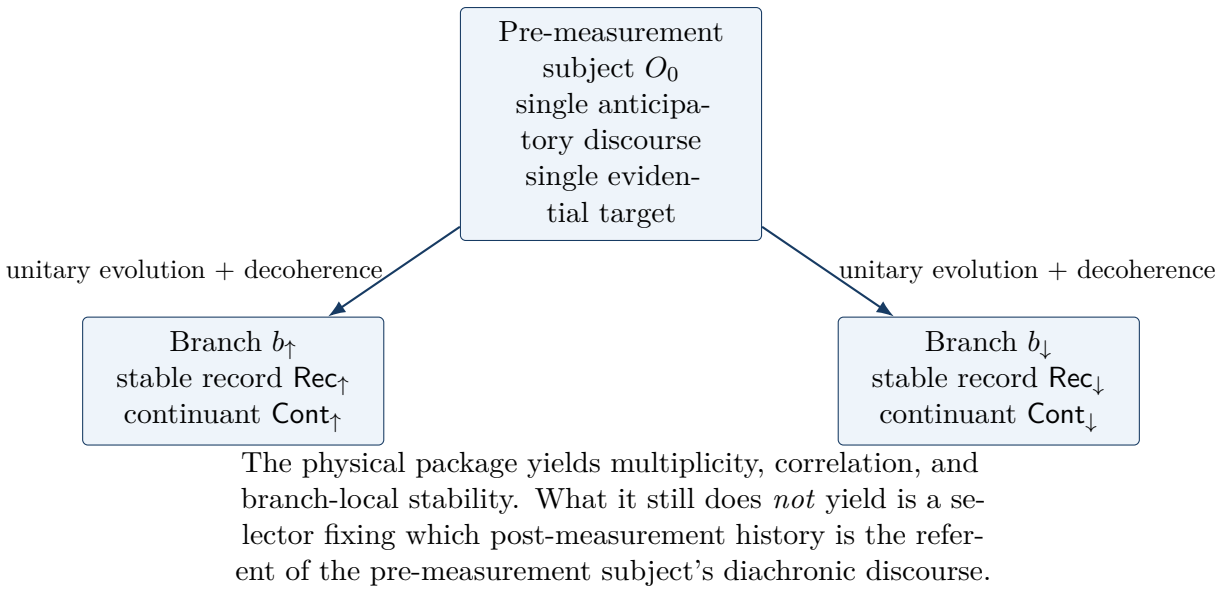

Figure 1: Minimal branching diagram for the theorem ladder. The Everettian base package yields multiple stable post-measurement records, but it does not physically select which history is the diachronic referent of the pre-measurement discourse.

4.2 Branch-relative determinacy versus diachronic reference

Once the Everettian base package is in place, one may grant a great deal. Each realized branch can contain a determinate local observer, a determinate local record, and a determinate local outcome. What still does not follow is that the pre-measurement subject's diachronic discourse has a uniquely fixed referent.

Definition 4.9 (Successor-qualified history)**.** A physically realized post-measurement history $b \in \mathcal{B}$ is *successor-qualified* for the purposes of a given piece of diachronic discourse if it contains both

1. a determinate measurement record $\mathsf{Rec}(b)$ appropriate to that discourse, and
2. a post-measurement continuant $\mathsf{Cont}(b)$ satisfying the continuity conditions presupposed by the discourse between the pre-measurement subject and the future subject.

Lemma 4.10 (Multiplicity of successor-qualified histories yields multiplicity of admissible assignments)**.** *Let $\mathcal{M}$ be a physically non-selective measurement episode and let d be a record-sensitive*

claim in its target discourse. If there exist distinct successor-qualified histories $b_i, b_j \in \mathcal{B}$ for d, then $\mathsf{ARef}(\mathcal{M})$ *contains at least two distinct admissible reference assignments relevant to d.*

Proof. Because b_i is successor-qualified for the discourse involving d, it satisfies the conditions that make the pair $(b_i, \mathsf{Rec}(b_i))$ eligible to serve as a referent for d. So there is an admissible assignment ρ_i sending d to $(b_i, \mathsf{Rec}(b_i))$. The same reasoning applies to b_j, yielding an admissible assignment ρ_j sending d to $(b_j, \mathsf{Rec}(b_j))$. Because $b_i \neq b_j$, the assignments are distinct. Therefore there are at least two admissible assignments relevant to d. □

Lemma 4.11 (Branch-relative determinacy does not entail diachronic determinacy)**.** *The existence of determinate branch-relative post-measurement experiences does not determine which such branch-relative experience is the referent of pre-measurement diachronic first-person discourse.*

Proof. Consider a branching model in which two post-measurement successors O_1 and O_2 each possess determinate branch-local experiences and stable branch-relative records. Now consider two admissible assignments: one taking O_1 to be the referent of the pre-measurement subject's diachronic discourse, the other taking O_2 to be that referent. The branch-local determinacy of each successor remains unchanged across the two assignments. What changes is only the diachronic successor-reference relation. Hence branch-local determinacy does not entail diachronic referential determinacy. □

Corollary 4.12 (Determinacy within branches is compatible with global referential underdetermination)**.** *A theory may deliver determinate experiences and records inside every realized branch while still failing to determine which branch-history is the referent of pre-measurement memory, anticipation, and evidential discourse.*

Proof. This is an immediate restatement of Theorem 4.11. The theorem establishes that local determinacy in each branch and diachronic determinacy across branches are distinct conditions. The first may hold while the second fails. □

Proposition 4.13 (Successor multiplicity marks a structural distinction)**.** *The difference between branch-local determinacy and diachronic referential determinacy marks a structural difference between two explanatory questions:*

1. *what records and observer-stages exist in each realized history; and*
2. *which such history is the referent of pre-measurement first-person and evidential discourse.*

Proof. The first question is answered by the branch-local facts contained in each history: what outcome is recorded there, what observer-stage exists there, and what local memory state exists there. The second question is answered only if the theory determines which realized history is relevant to the pre-measurement subject's diachronic claims. A theory may answer the first question for every branch and still leave the second unanswered. That possibility is exactly what Theorem 4.11 shows. Therefore the distinction is structural. □

4.3 Decoherence and remaining indeterminacy

All discussion of evidential and experiential reference in this section is conditional on the target explanatory burden fixed in Section 2. The issue is not whether decoherence is useful or real. The issue is whether decoherence, by itself, fixes diachronic reference.

Definition 4.14 (Record stability under decoherence)**.** A measurement episode $\mathcal{M}$ satisfies *decoherence-based record stability* if for every physically realized history $b \in \mathcal{B}$, the record $\mathsf{Rec}(b)$ is dynamically stable enough to support branch-local memory and observational reliability, and the corresponding branch remains approximately non-interfering with the others for the purposes of the episode under discussion.

Proposition 4.15 (Decoherence stabilizes records without selecting referents)**.** *Let $\mathcal{M}$ be a physically non-selective measurement episode satisfying decoherence-based record stability. Then decoherence stabilizes the records contained in each $b \in \mathcal{B}$ but does not by itself privilege one b as the unique referent of pre-measurement diachronic discourse.*

Proof. By decoherence-based record stability, each history $b \in \mathcal{B}$ contains a dynamically stable record $\mathsf{Rec}(b)$ that may support branch-local memory and branch-local observational discourse. This is the positive achievement of decoherence.

To show that decoherence does not by itself privilege a unique diachronic referent, suppose for contradiction that it does. Then decoherence would provide a selector: it would reduce the admissible reference possibilities relevant to the discourse to exactly one privileged assignment. But Decoherence-based record stability, as defined here, supplies only local stability and effective non-interference of the realized histories. It says that each realized branch carries a stable record; it does not say that one stable branch is physically privileged over the others as *the* continuation relevant to pre-measurement discourse. Adding such a privilege would amount to adding a selector not already contained in the bare statement of decoherence-based stability. Therefore decoherence stabilizes records without by itself selecting referents. □

Theorem 4.16 (Decoherence non-selection theorem)**.** *If a theory's post-measurement contribution to a measurement episode consists only of the physical realization of multiple branch-histories together with decoherence-based record stability, then the theory does not by that contribution alone fix diachronic reference for measurement records.*

Proof. Assume the theory's post-measurement contribution consists only of the physical realization of multiple histories together with decoherence-based record stability. By Proposition 4.15, that contribution stabilizes the records in each realized history but does not itself provide a selector. If the episode is physically non-selective, then at least two admissible reference assignments remain available. Therefore the theory's post-measurement contribution, as specified, does not fix diachronic reference for the measurement records. □

4.4 Centered descriptions, self-location, and decision theory

The most common way of resisting the previous result is to say that Everettian theory has resources for reasoning under multiplicity: centered-world semantics, self-locating uncertainty, and decision-theoretic branch weighting [7–9]. The present claim is not that such resources are incoherent. The claim is that they presuppose a prior typing of successors.

Definition 4.17 (Centered continuation description)**.** A *centered continuation description* for a measurement episode $\mathcal{M}$ is a pair (b, c_b) where $b \in \mathcal{B}$ is a physically realized post-measurement history and c_b is a center-marker intended to treat the post-measurement continuant in b as a candidate continuation of the pre-measurement subject for first-person or evidential discourse.

Definition 4.18 (Centered enrichment)**.** A *centered enrichment* of a physically non-selective measurement episode $\mathcal{M}$ is the passage from the uncentered history family $\mathcal{B}$ to a family of centered continuation descriptions

$$\mathcal{C}_M = \{(b, c_b) : b \in \mathcal{B}\}.$$

The enrichment is *selector-free* if it adds no rule that privileges exactly one center (b, c_b) as the referent of pre-measurement diachronic discourse.

Lemma 4.19 (Centered enrichment preserves multiplicity)**.** *Let $\mathcal{M}$ be a physically non-selective measurement episode and let $\mathcal{C}_M$ be a selector-free centered enrichment of $\mathcal{M}$. Then if $|\mathcal{B}| \geq 2$, the centered enrichment contains at least two distinct candidate centers.*

Proof. Because $\mathcal{M}$ is physically non-selective, $|\mathcal{B}| \geq 2$. So there exist distinct histories $b_i \neq b_j$ in $\mathcal{B}$. By definition of centered enrichment, both give rise to centered continuation descriptions (b_i, c_{b_i}) and (b_j, c_{b_j}). Since the underlying histories are distinct, these centered descriptions are distinct as well. Therefore the centered enrichment contains at least two distinct candidate centers. □

Proposition 4.20 (Centered enrichment does not by itself select a referent)**.** *A selector-free centered enrichment of a physically non-selective measurement episode does not by itself determine which centered continuation description is the referent of pre-measurement diachronic measurement discourse.*

Proof. By Lemma 4.19, selector-free centered enrichment preserves multiplicity of candidate centers. To determine which centered continuation description is the referent of pre-measurement discourse, one would need a rule that singles out one center from the enriched set $\mathcal{C}_M$. But selector-free centered enrichment, by definition, adds no such rule. So the centered enrichment redescribes the multiplicity of candidate continuants; it does not eliminate that multiplicity or privilege one member of it as the unique referent. □

Definition 4.21 (Typed successor set)**.** A set Σ_M of post-measurement continuants is a *typed successor set* for a piece of diachronic measurement discourse iff each member of Σ_M is already fixed as an admissible candidate continuation of the pre-measurement subject for the truth conditions of that discourse.

Lemma 4.22 (Self-locating uncertainty requires a typed successor set)**.** *A self-locating uncertainty rule can guide first-person credence only if it ranges over a typed successor set.*

Proof. A self-locating uncertainty rule distributes credence across candidate future centers or successors. For that distribution to be first-personally meaningful, the rule's domain must already consist of the candidates relevant to the subject whose uncertainty is in question. If the domain were not already typed in that way, the rule would distribute credence across a merely physical collection of post-measurement continuants without having fixed that these are the continuants relevant to the present subject's anticipatory discourse. In that case the rule would fail to determine the truth conditions of first-person claims such as "I will observe o". So self-locating uncertainty requires a typed successor set. □

Definition 4.23 (Decision-theoretic branching rule)**.** A *decision-theoretic branching rule* for a measurement episode $\mathcal{M}$ is any rule that assigns utilities, preferences, or credences across a set of

candidate post-measurement continuants or records and is then used to guide pre-measurement deliberation.

Lemma 4.24 (Decision rules require typed successor events)**.** *A decision-theoretic branching rule can guide action only if the successor events or records to which it assigns utilities or credences are already typed as the relevant successors for the discourse being used by the agent.*

Proof. A decision-theoretic rule guides action only if its outputs bear on what the present agent should expect, prefer, or do. To have that bearing, the successor events or records over which the rule ranges must already be the successor events or records relevant to the present agent. If those items are not already typed as the relevant successors for the discourse in which the decision is being made, then the rule's outputs are not yet action-guiding in the intended first-person sense. The rule would at best distribute weights over an untyped collection of physical continuants without thereby determining which of them are the future experiences or records to which the present agent's claims refer. Therefore the decision rule requires successor events or records that are already referentially typed for the discourse in question. □

Proposition 4.25 (Weights over branches are not selectors)**.** *Assigning amplitudes, credences, utilities, or caring measures to realized branches does not by itself fix which history is the referent of pre-measurement diachronic discourse.*

Proof. A selector determines which realized history is the relevant referent. A weight assignment instead orders or measures the candidate histories while leaving the domain of candidates in place. Unless one already knows what the weighted items are the candidates *for*, the weights do not fix which history the pre-measurement discourse is about. They therefore presuppose rather than provide the needed referential typing. □

Theorem 4.26 (Everettian downstream theorem)**.** *Centered descriptions, self-locating uncertainty, and decision-theoretic branch-weighting are downstream of the Everettian non-selection problem. They presuppose successor typing and therefore cannot by themselves supply the diachronic reference fixation whose absence constitutes the problem identified in this paper.*

Proof. By Proposition 4.20, centered enrichment without a selector preserves rather than eliminates multiplicity of candidate continuants. By Lemma 4.22, any self-locating treatment can guide first-person credence only if its domain is already a typed successor set. By Lemma 4.24, a decision-theoretic branching rule can guide action only if its successor events are already typed as the relevant successors for the present agent's discourse. But Theorem 4.8 and Theorem 4.11 show that the bare Everettian physical package leaves multiple realized histories in play without fixing which one is the relevant diachronic referent. Therefore centered descriptions, self-location, and decision theory begin only after the missing referential work has been done. They are downstream of the non-selection problem rather than solutions to it. □

Section 4 has established that the standard Everettian physical package yields multiple stable branch-local records and no unique diachronic referent, and that self-location and decision-theoretic replies presuppose rather than remove that problem. The next section turns to how measurement records function as evidential anchors and what happens to theory-level confirmation when admissible reference assignments diverge.

5 Measurement records as evidential anchors and the structure of theory-level confirmation

Section 3 isolated the quotient structure of admissible reference assignments and identified fixation commutation as the bridge invariant. The present section uses that bridge to turn the quotient into a measurement theorem about evidence. Measurement records are evidential anchors; once that role is explicit, divergence among admissible quotient classes is no longer a semantic curiosity but a question about whether the theory fixes its own evidential output.

Central hinge. The quotient theorem identifies which divergences among admissible reference assignments are standing-bearing, and Section 3 proves that fixation commutation is the bridge invariant for theory-level evidential use. The present section therefore asks a narrower question: when quotient-relevant divergence survives into comparative evidential roles, does the theory fix its own evidential output?

5.1 Measurement records as evidential anchors

Definition 5.1 (Evidential anchor)**.** A measurement record $r \in \mathcal{R}$ is an *evidential anchor* for a diachronic measurement claim $d \in \mathcal{D}$ just in case the theory-level use of d depends on taking r as the referent of d and then using the induced evidential proposition $\mathsf{Prop}_\rho(d)$ to compare rival hypotheses in $\mathcal{H}$.

Definition 5.2 (Theory-level confirmation)**.** A *theory-level confirmation verdict* is a comparative confirmational verdict attributable to the theory on the fixed scope rather than to an unresolved choice among admissible reference assignments. Equivalently, a theory-level confirmation verdict is well-defined only if the theory's admissible use does not leave the comparative evidential role of its measurement records unsettled across admissible standing-equivalence classes.

Definition 5.3 (Comparative hypothesis ordering induced by a record)**.** Let $d \in \mathcal{D}$ be an evidential claim and let $\rho \in \mathsf{ARef}$ be an admissible reference assignment. The *comparative hypothesis ordering induced by d under ρ* is the function

$$\mathsf{Ord}^d_\rho(H_i, H_j) := \mathsf{CR}(\mathsf{Prop}_\rho(d))(H_i, H_j) \qquad (H_i, H_j \in \mathcal{H}).$$

A theory-level confirmation claim built from d is theory-relative only if this ordering is invariant on the admissible quotient classes relevant to d.

Proposition 5.4 (Evidential use of records is assignment-sensitive)**.** *Let $d \in \mathcal{D}$ be a diachronic measurement claim used as an evidential anchor. Then every admissible reference assignment $\rho \in \mathsf{ARef}$ determines a corresponding comparative hypothesis ordering Ord^d_ρ on $\mathcal{H}$. Therefore any unresolved variation among admissible quotient classes is already variation in evidential use unless all such orderings coincide.*

Proof. By definition of admissible reference assignment, $\rho(d)$ is a physically realized and admissible branch-record pair for d. By definition of the induced evidential proposition, $\rho(d)$ determines $\mathsf{Prop}_\rho(d) \in \mathcal{E}$. By definition of confirmational role, $\mathsf{CR}(\mathsf{Prop}_\rho(d))$ is a function on ordered pairs

of hypotheses. So Ord^d_ρ is well-defined for every admissible assignment. If different admissible quotient classes induce different orderings, then the record is doing different comparative evidential work under those classes. Hence unresolved variation among admissible quotient classes is already variation in evidential use unless all orderings coincide. □

5.2 The semantic hinge

Lemma 5.5 (Two-case exhaustion for evidential reference)**.** *Let $d \in \mathcal{D}$ be used evidentially. Then exactly one of the following holds:*

1. *all admissible reference assignments lie in a single $\approx_{\mathrm{MR}}$-class on the evidential use of d; or*
2. *there exist admissible reference assignments whose evidential values for d lie in distinct $\approx_{\mathrm{MR}}$-classes.*

No third same-scope case exists.

Proof. Either the admissible assignments agree on the standing-bearing evidential value of d, or they do not. If they do, then they lie in one $\approx_{\mathrm{MR}}$-class on that use. If they do not, then there exist at least two distinct $\approx_{\mathrm{MR}}$-classes. By Definition 2.1 and Proposition 2.9, no third same-scope standing-bearing locus remains. □

Theorem 5.6 (No auxiliary parameterization of theory-level confirmation)**.** *Let T be a theory and let ARef be the class of its admissible reference assignments for the fixed measurement setting. If comparative confirmational verdicts for T vary across admissible $\approx_{\mathrm{MR}}$-classes, then those verdicts are not attributable to T as such. They are attributable only to the pair consisting of T together with the selected quotient class. Consequently, such verdicts do not constitute theory-level confirmation in the sense relevant to the target explanatory practice.*

Proof. By Definition 5.2, a theory-level confirmation verdict must be attributable to the theory on the fixed scope rather than to a further unresolved choice among admissible reference assignments. Suppose instead that distinct admissible $\approx_{\mathrm{MR}}$-classes yield different comparative orderings for some evidential claim d. Then the comparative support relation between hypotheses is not fixed by T alone. It varies with the chosen admissible quotient class. In that case the verdict is attributable only to $(T, [\rho])$ for some chosen class $[\rho] \in \mathsf{ARef}/\approx_{\mathrm{MR}}$, not to T as such. Such verdicts are therefore assignment-parameterized rather than theory-level. □

Theorem 5.7 (Theory-level confirmation criterion)**.** *For an evidential claim $d \in \mathcal{D}$, theory-level confirmation is well-defined if and only if the map*

$$[\rho] \longmapsto \mathsf{Ord}^d_\rho$$

from $\mathsf{ARef}/\approx_{\mathrm{MR}}$ to comparative hypothesis orderings is constant. Equivalently, if there exist admissible assignments whose comparative confirmational roles differ, then no theory-level confirmation verdict is fixed by the theory as such.

Proof. If the displayed map is constant, then all admissible assignments induce the same comparative ordering for d, so the verdict is attributable to the theory on the fixed scope rather than to a selected admissible quotient class. Conversely, if the map is not constant, then there exist admissible

assignments inducing different comparative orderings. By Corollary 3.18, that divergence is quotient-relevant and therefore standing-bearing rather than semantic skin. By Theorem 5.6, the resulting verdict is attributable only to the pair consisting of the theory together with a chosen admissible quotient class. Hence no theory-level confirmation verdict is fixed by the theory as such. □

Corollary 5.8 (Confirmational consequence)**.** *If there exist admissible reference assignments that differ in comparative confirmational role, then confirmation is assignment-relative rather than theory-relative on the fixed scope.*

Proof. Immediate from Theorem 5.7. □

5.3 Fixation as the bridge invariant

Definitions 3.19 and 3.20, together with Theorem 3.22 and Corollary 3.23, were established already in Section 3. They are placed there because fixation is the bridge invariant between quotient structure and all later applications. The present section uses that bridge to state the confirmation criterion and the stabilization consequence for physically non-selective theories.

Theorem 5.9 (Reference-stabilization necessity)**.** *If a physically non-selective theory is used to underwrite theory-level confirmation and truth-apt diachronic discourse whose measurement records function as evidential anchors, then the theory or its admissible interpretive completion must restrict admissible reference assignments strongly enough that fixation commutes with admissible redescription on the target discourse.*

Proof. By Lemma 5.5, admissible assignments either collapse into a single standing-bearing class on the target discourse or they do not. If they do not, then by Theorem 5.7 and Theorem 3.22, theory-level confirmation through measurement records is undefined on the fixed scope. But the standing conditional of Section 2 assumes that the theory is being used to underwrite theory-level confirmation and truth-apt diachronic discourse in which measurement records function as evidential anchors. Therefore the role-divergent case is incompatible with the theory's intended explanatory use. The only remaining option is that the theory or its admissible interpretive completion restricts admissible reference assignments strongly enough for fixation to commute with admissible redescription on the target discourse. □

Proposition 5.10 (Selector and semantic-infrastructure normal form)**.** *Any successful stabilization strategy for admissible measurement-record reference takes exactly one of two normal forms:*

1. selector form*, in which singular reference is fixed by selecting one history or one successor-relation as privileged; or*

2. semantic-infrastructure form*, in which admissible assignments remain multiple but are restricted strongly enough to collapse into one standing-equivalence class on the target discourse.*

No third same-scope stabilization form remains.

Proof. If a successful stabilization strategy fixes a unique referent outright, it has selector form. If it does not fix a unique referent outright but still preserves theory-level confirmation, then by Theorem 5.9 it must restrict admissible assignments strongly enough that fixation commutes and the assignments collapse into one standing-equivalence class on the target discourse. That is

semantic-infrastructure form. By Proposition 2.9, no third same-scope locus of admissibility-bearing work remains. □

Proposition 5.11 (Scope of the main theorem)**.** *Theorems 5.7 and 5.9 concern theory-level confirmation on the fixed scope. They do not deny that branch-internal discourse may be perfectly determinate once a branch is taken as fixed.*

Proof. The hypotheses of Definitions 5.2 and 3.20 explicitly concern theory-level evidential use on the fixed downstream substrate. Nothing in those definitions denies local branch-internal determinacy once a particular branch-history is held fixed. The theorems only state that such local determinacy does not by itself suffice for theory-level confirmation on the fixed scope. □

6 A toy model of confirmational divergence

Abstract theorems are necessary, but here a concrete toy model helps show exactly what the problem looks like in practice. The goal is not to model a realistic laboratory in every detail. The goal is to exhibit, in the smallest possible setting, how a single branching measurement episode can generate opposed theory-level confirmation claims when admissible reference assignments are left unresolved.

6.1 The model

Definition 6.1 (Toy branching confirmation model)**.** A *toy branching confirmation model* is a measurement scenario $\mathcal{M}_{\text{toy}}$ with the following data:

1. two rival hypotheses $\mathcal{H} = \{H_U, H_D\}$;
2. two physically realized post-measurement histories $\mathcal{B} = \{b_U, b_D\}$;
3. two corresponding branch records $r_U := \mathsf{Rec}(b_U)$ and $r_D := \mathsf{Rec}(b_D)$;
4. a single diachronic claim $d \in \mathcal{D}$ whose intended English gloss is "this is the measurement record I am using as evidence";
5. two admissible reference assignments $\rho_U, \rho_D \in \mathsf{ARef}(\mathcal{M}_{\text{toy}})$ with
$$\rho_U(d) = (b_U, r_U), \qquad \rho_D(d) = (b_D, r_D);$$
6. induced evidential propositions $E_U := \mathsf{Prop}_{\rho_U}(d)$ and $E_D := \mathsf{Prop}_{\rho_D}(d)$ with confirmational roles satisfying
$$\mathsf{CR}(E_U)(H_U, H_D) = 1, \qquad \mathsf{CR}(E_D)(H_U, H_D) = -1.$$

Intuitively, E_U is a record that favors H_U over H_D, while E_D favors H_D over H_U. The physical non-selectivity of the scenario is represented by the fact that both branch-record pairs are realized and both are admissible assignments for the same diachronic claim d.

Proposition 6.2 (Hypothesis-order instability witness)**.** *Let H_U, H_D be rival hypotheses, let d be the fixed measurement claim in a toy branching confirmation model, and let ρ_U, ρ_D be admissible reference assignments mapping that same pre-measurement discourse to realized records r_U, r_D.*

If under ρ_U the induced evidential proposition favors H_U over H_D, while under ρ_D the induced evidential proposition favors H_D over H_U, then comparative confirmation is not fixed by the theory alone.

Proof. By Definition 6.1, ρ_U and ρ_D are admissible reference assignments for the same diachronic claim d, and

$$\mathsf{CR}(\mathsf{Prop}_{\rho_U}(d))(H_U, H_D) = 1, \qquad \mathsf{CR}(\mathsf{Prop}_{\rho_D}(d))(H_U, H_D) = -1.$$

So under ρ_U the record referred to by d favors H_U over H_D, while under ρ_D the record referred to by d favors H_D over H_U. The comparative ordering of the hypotheses therefore depends on which admissible assignment is used. Hence comparative confirmation is not fixed by the theory alone. □

Theorem 6.3 (Hypothesis-order reversal in the toy model)**.** *In every toy branching confirmation model, the comparative ordering of H_U and H_D reverses across admissible reference assignments. Consequently, the theory-level confirmation claim built from d is reference-relative rather than theory-relative.*

Proof. By Definition 6.1,

$$\mathsf{Ord}^d_{\rho_U}(H_U, H_D) = \mathsf{CR}(E_U)(H_U, H_D) = 1$$

and

$$\mathsf{Ord}^d_{\rho_D}(H_U, H_D) = \mathsf{CR}(E_D)(H_U, H_D) = -1.$$

So under ρ_U, the relevant record favors H_U over H_D, whereas under ρ_D, the relevant record favors H_D over H_U. The comparative ordering is therefore reversed across admissible assignments.

Suppose the theory-level confirmation claim built from d were theory-relative. Then, by Definition 5.2 and Theorem 5.7, the induced comparative ordering would have to be invariant across admissible assignments. But it is not. Therefore the claim is reference-relative rather than theory-relative. □

Corollary 6.4 (The same physical branching episode can support opposed confirmation claims)**.** *In the toy model, the same physical branching episode supports both of the following claims under different admissible assignments:*

"the record referred to by d confirms H_U over H_D" and "the record referred to by d confirms H_D over H_U".

Proof. The first claim is true under ρ_U because $\mathsf{Ord}^d_{\rho_U}(H_U, H_D) = 1$. The second claim is true under ρ_D because $\mathsf{Ord}^d_{\rho_D}(H_U, H_D) = -1$, which is equivalent to saying that the record favors H_D over H_U. By Theorem 6.3, both verdicts arise from the same branching episode under different admissible assignments. □

Remark 6.5. Table 1 is not a replacement for the theorem ladder; it is a visibility aid. The point of the toy model is to make plain that the problem is not some exotic semantic subtlety. If admissible assignments are allowed to diverge in confirmational role, the same physical episode can be used to support opposed claims about which hypothesis the evidence favors.

Assignment	Record referred to by d	Induced evidential proposition	Comparative role on (H_U, H_D)
ρ_U	(b_U, r_U)	E_U	favors H_U over H_D
ρ_D	(b_D, r_D)	E_D	favors H_D over H_U

Table 1: The toy model makes the danger concrete: the same branching measurement episode supports opposite comparative hypothesis orderings when admissible reference assignments remain unresolved.

7 Diachronic practice: memory, anticipation, and agency inherit the same lineage

The point of the measurement-reference quotient is not confined to evidential comparison. The same records support memory claims, anticipatory claims, and deliberative guidance. If those practices are treated as truth-apt on the fixed scope, then they must all inherit a common referential lineage. This section makes that dependence explicit.

7.1 Memory and anticipation

Definition 7.1 (Memory claim)**.** A *memory claim* is a diachronic claim whose truth conditions depend on taking a present subject-stage to stand in the relevant same-scope lineage relation to a past record-bearing history.

Definition 7.2 (Anticipation claim)**.** An *anticipation claim* is a diachronic claim whose truth conditions depend on taking a present subject-stage to stand in the relevant same-scope lineage relation to one or more future record-bearing histories.

Definition 7.3 (Referential lineage)**.** For an admissible reference assignment ρ, the *referential lineage* induced by ρ is the same-scope chain of subject-stage, successor-typing, and record-reference relations through which the claims in $\mathcal{D}$ receive their standing-bearing values under ρ.

Definition 7.4 (Diachronic claim)**.** A *diachronic claim* is any claim in the target explanatory practice whose standing-bearing value depends on locating a present subject-stage, a past or future measurement record, or both within the same referential lineage.

Theorem 7.5 (Diachronic closure)**.** *Any admissible diachronic claim factors through the same standing-preserving equivalence relation on* ARef*, namely the measurement-reference quotient* $\mathsf{ARef}/\approx_{\mathrm{MR}}$*. Equivalently, there is no admissible diachronic operator whose standing-bearing content is independent of that quotient.*

Proof. By Definition 7.4, a diachronic claim belongs to the target explanatory practice only insofar as it receives a standing-bearing value from the same record-sensitive profile map Prof used to define $\approx_{\mathrm{MR}}$. By Theorem 3.12, every standing-bearing content space for admissible measurement-reference assignments factors through the unique quotient $\mathsf{ARef}/\approx_{\mathrm{MR}}$. So if a diachronic claim failed to factor through that quotient, two assignments in the same $\approx_{\mathrm{MR}}$-class would have to yield different standing-bearing values for the claim, contradicting the definition of $\approx_{\mathrm{MR}}$ and Theorem 3.14. Hence any admissible diachronic claim factors through the same standing-preserving equivalence relation. □

Lemma 7.6 (Memory truth conditions depend on record reference)**.** *If two admissible assignments differ on which past record is the referent of a memory claim, then they need not assign the same truth value to that claim.*

Proof. A memory claim is true only if the present subject-stage is related, under the relevant diachronic lineage, to the past record the claim is about. Changing which record is taken as the referent can therefore change whether the claim is true. Hence memory truth conditions depend on record reference. □

Lemma 7.7 (Anticipation truth conditions depend on successor typing)**.** *If two admissible assignments differ on which future record-bearing history is the referent relevant to an anticipation claim, then they need not assign the same truth value to that claim.*

Proof. An anticipation claim attributes a future outcome to the present subject-stage. If admissible assignments differ on which future record-bearing history is the relevant continuation, then they may differ on whether the attributed outcome occurs in that history. So anticipation truth conditions depend on successor typing and record reference. □

7.2 Agency and deliberation

Definition 7.8 (Agential deliberation claim)**.** An *agential deliberation claim* is a claim whose standing-bearing use guides present action by appeal to future measurement-record outcomes treated as belonging to the same diachronic practice-package.

Lemma 7.9 (Agential guidance requires typed future records)**.** *If a deliberative claim is to guide present action by appeal to future measurement outcomes, then the future records appealed to must be typed as belonging to the same referential lineage as the present subject-stage.*

Proof. Deliberation links a present act to future consequences for the same diachronic practice. If the future records appealed to are not typed as belonging to the relevant lineage, then the deliberative claim has no determinate target for guidance. So agential guidance requires typed future records. □

Theorem 7.10 (Decision-theoretic dependence for literal first-person anticipation)**.** *Any literal first-person decision-theoretic use of branching quantum theory presupposes prior successor typing and record reference. It does not create them.*

Proof. A decision-theoretic rule can distribute weights or preferences only over already identified future alternatives. In the present setting those alternatives are future record-bearing histories. If successor relations and the relevant records are not already typed, then the decision rule lacks a determinate space of future options to which its weights or preferences apply. So decision-theoretic use presupposes rather than creates reference fixation. □

Proposition 7.11 (Branch-aggregative retyping)**.** *A framework may avoid literal first-person anticipation by retyping deliberation into branch-aggregative or impersonal terms. Such a framework does not remove the need for referential typing; it changes the discourse whose truth conditions are at issue.*

Proof. If a framework replaces literal first-person anticipation by impersonal aggregation over branches, then it no longer uses the same diachronic first-person claims as the target explanatory practice. The move therefore avoids the strongest lineage requirement only by changing the discourse at issue. That is retyping, not elimination of the prior dependence. □

7.3 A unified lineage theorem

Theorem 7.12 (Identity-preserving lineage theorem)**.** *On the fixed downstream substrate, memory, anticipation, deliberation, and theory-level confirmation all presuppose a fixed identity-preserving referential lineage through the measurement records used in those practices.*

Proof. By Lemma 7.6, memory claims depend on which past records are the relevant referents. By Lemma 7.7, anticipation claims depend on successor typing and future record reference. By Lemma 7.9, agential guidance depends on typed future records. By Proposition 5.4 and Theorem 5.7, theory-level confirmation depends on the standing-bearing evidential use of the same records. These four practices therefore do not rest on four independent structures. They rest on one identity-preserving referential lineage through the relevant measurement records. □

Theorem 7.13 (Diachronic inheritance theorem)**.** *On the fixed scope, memory, anticipation, deliberation, and theory-level confirmation all factor through the same measurement-reference quotient. Consequently, any quotient-relevant divergence in lineage assignment is inherited by all four practices together rather than by one of them in isolation.*

Proof. By Theorem 7.5, every admissible diachronic claim factors through the unique quotient $\mathsf{ARef}/\approx_{\mathrm{MR}}$; this is a special case of the general measurement-reference quotient theorem. Memory and anticipation claims depend on which record-bearing histories are taken as their referents (Lemmas 7.6 and 7.7); agential deliberation depends on the same typed future records (Lemma 7.9); and theory-level confirmation depends on the standing-bearing evidential use of those records (Theorem 5.7). Therefore all four practices factor through the same measurement-reference quotient. If divergence is quotient-relevant, then it is inherited by each practice through that shared factorization. □

Theorem 7.14 (Diachronic practice dependence theorem)**.** *If admissible reference assignments remain divergent on the fixed scope in a way that changes the referential lineage of the relevant measurement records, then memory, anticipation, deliberation, and theory-level confirmation become assignment-relative together. No one of the four can be stabilized in isolation while the shared quotient-level lineage remains divergent.*

Proof. By Theorem 7.12, the four practices share a common referential lineage. If admissible assignments diverge in a way that changes that lineage, then by Lemmas 7.6 and 7.7 they can assign different truth conditions to memory and anticipation claims; by Lemma 7.9 they can change the future records relevant to deliberation; and by Corollary 5.8 they can change theory-level confirmation. Since the dependence is shared, the instability is shared. The four practices therefore become assignment-relative together. □

Corollary 7.15 (Structural priority over probabilistic treatment)**.** *Questions about branch-weighting, credence, or rational choice are downstream of the lineage issue. They presuppose, rather than settle, the fixed referential lineage required by the target explanatory practice.*

Proof. Immediate from Theorems 7.10 and 7.12. A probabilistic or decision-theoretic treatment can operate only once the relevant future records and successor relations have already been typed into a fixed lineage. □

8 Interpretation taxonomy: where is singular reference fixed?

The previous sections have now done enough work that the interpretation discussion can be stated as a same-scope exhaustion theorem. The relevant question is not simply which interpretation one prefers. It is this: *where, if anywhere, is the work of fixing singular reference done strongly enough for measurement records to function as theory-level evidence on the fixed scope?*

8.1 Interpretation packages and the loci of reference-fixing work

Definition 8.1 (Interpretation package)**.** An *interpretation package* for a measurement scenario $\mathcal{M}$ is a triple

$$\mathfrak{I} = (T, \Lambda, \mathcal{D}^*),$$

where:

1. $T = (\Omega, \Delta)$ is the physical theory, with ontology or state-description Ω and dynamics Δ;
2. Λ is the class of further resources appealed to in fixing the referents of measurement discourse, including added ontology, semantic restrictions, or discourse-level policies; and
3. $\mathcal{D}^*$ is the class of diachronic measurement claims the package is intended to underwrite.

Definition 8.2 (Discourse-committing and discourse-deflationary packages)**.** An interpretation package $\mathfrak{I}$ is *discourse-committing* if it takes the claims in $\mathcal{D}^*$ to be truth-apt and confirmation-guiding through the measurement records of the scenario. It is *discourse-deflationary* if it does not take the theory to underwrite those claims in the required truth-apt diachronic sense.

Definition 8.3 (Dynamic selection)**.** A discourse-committing package realizes *dynamic selection* if singular reference is fixed by the physical evolution law itself: for the relevant measurement episode, the dynamics Δ alone suppresses or removes outcome multiplicity strongly enough that exactly one post-measurement history remains eligible as the referent relevant to $\mathcal{D}^*$.

Definition 8.4 (Ontic privileging)**.** A discourse-committing package realizes *ontic privileging* if singular reference is fixed by additional ontological structure not determined by the bare dynamics alone. Equivalently, the physical description still requires an extra state variable, trajectory, configuration, or comparable ontic fact to designate which history is the actual referent.

Definition 8.5 (Semantic restriction)**.** A discourse-committing package realizes *semantic restriction* if the physical theory remains non-selective, but admissible reference assignments are restricted strongly enough that the surviving assignments form a single $\approx_{\mathrm{MR}}$-class on the target discourse.

8.2 From stabilization to structural exhaustion

Lemma 8.6 (Discourse commitment forces stabilization)**.** *If an interpretation package $\mathfrak{I}$ is discourse-committing, then it must stabilize admissible reference assignments strongly enough to avoid the role-divergent case of Lemma 5.5.*

Proof. A discourse-committing package takes theory-level confirmation, memory, anticipation, deliberation, or some combination of them to be truth-apt and guided by measurement records. By Theorem 5.9, any such use requires admissible reference assignments to collapse strongly enough that fixation commutes with admissible redescription on the target discourse. So discourse commitment forces stabilization. □

Definition 8.7 (Structural locus of singular-reference fixing)**.** A same-scope resource that purports to fix singular reference does its standing-bearing work at exactly one of the following loci:

1. **boundary locus**: by acting as an admissibility-relevant selector or parameter at the pre-admissible boundary;
2. **quotient locus**: by refining the quotient-level standing-bearing content itself through a new invariant or privileged continuation relation; or
3. **representation locus**: by restricting how already-admissible representatives or uses are allowed to count without changing the quotient-level physical package.

Theorem 8.8 (Structural locus trichotomy)**.** *Any admissibility-preserving same-scope selector must operate at exactly one of the loci in Definition 8.7. No fourth same-scope locus exists.*

Proof. By Proposition 2.9, any same-scope admissibility-bearing difference must appear either in standing classification or in licensed continuation. The fixed downstream substrate closes any second same-scope invariant except as bookkeeping or scope change, and the role-decomposition results imported with that substrate leave no fourth admissibility-bearing role beside anchor-level, quotient/tensor-level, and representational/skin-level work. A selector that acts before quotient-level content is fixed therefore acts at the boundary; a selector that changes standing-bearing content acts on the quotient; and a selector that leaves quotient content untouched but changes how representatives may count acts via representation. If it does none of these, it is bookkeeping only and not a selector. So no fourth same-scope locus remains. □

Corollary 8.9 (Boundary selectors are forbidden)**.** *If a same-scope selector operates at the boundary locus, it introduces admissibility-relevant parameterization or selection and is therefore forbidden on the fixed scope.*

Proof. By Theorem 8.8, a boundary-locus selector acts before quotient-level content is fixed. But Theorem 2.8 and Lemma 2.7 exclude same-scope admissibility- relevant parameterization or selection at the boundary. So a boundary selector is forbidden. □

Corollary 8.10 (Quotient refinements force new invariant structure)**.** *If a same-scope selector operates at the quotient locus, then it refines standing-bearing content by a new invariant or privileged continuation relation. On the fixed scope that is not an additional neutral interpretation layer; it is a scope change relative to the non-selective base package.*

Proof. A quotient-locus selector changes which standing-bearing content counts as the relevant continuation. By Theorem 3.14, no alternative same-scope admissibility-preserving quotient survives beside $\approx_{\mathrm{MR}}$. So any quotient refinement either coincides with the existing quotient or introduces a new same-scope invariant. The latter is excluded by the fixed downstream substrate except as scope change. □

Corollary 8.11 (Representation restrictions yield semantic restriction or deflation)**.** *If a same-scope selector operates only at the representation locus while leaving the physical package non-selective, then either it restricts admissible representatives to one quotient class (semantic restriction) or it refuses the target truth-apt discourse (discourse deflation).*

Proof. A purely representational selector cannot change boundary status or quotient-level content without leaving the representation locus. If the target discourse remains truth-apt and theory-relative, the only way such a selector can matter is by restricting admissible representatives strongly enough that their use collapses into one quotient class; that is semantic restriction. If the target discourse is instead refused, the package is discourse-deflationary. No third representation-locus option remains. □

Theorem 8.12 (Selector-exhaustion theorem)**.** *Any admissibility-preserving selector must do exactly one of the following:*

1. *introduce boundary parameterization or a boundary selector, which is forbidden;*
2. *refine standing-bearing quotient content by a new invariant or privileged continuation relation, which is a scope change on the fixed substrate;*
3. *restrict representation so that admissible assignments collapse into one quotient class, which is semantic restriction; or*
4. *refuse the target truth-apt discourse, which is discourse deflation.*

No fifth same-scope selector route exists.

Proof. By Theorem 8.8, every same-scope selector operates at the boundary, on the quotient, or via representation. Boundary-locus selectors are forbidden by Corollary 8.9. Quotient-locus selectors refine standing-bearing content and therefore introduce a new invariant or privileged continuation relation; by Corollary 8.10 this is a scope change on the fixed substrate. Representation-locus selectors fall under Corollary 8.11: they are either semantic restriction or discourse deflation. These cases are exhaustive, so no fifth same-scope selector route remains. □

Theorem 8.13 (Interpretation taxonomy)**.** *Every interpretation package $\mathfrak{I}$ for a measurement scenario falls into exactly one of the following structural classes:*

1. ***Dynamic selection.*** *Singular reference is fixed by physical dynamics.*
2. ***Ontic privileging.*** *Singular reference is fixed by added ontological structure or a privileged trajectory.*
3. ***Semantic restriction.*** *The physical theory remains non-selective, but admissible reference assignments are restricted to a single standing-equivalence class.*
4. ***Discourse deflation.*** *The package does not take the theory to underwrite the target class of truth-apt diachronic measurement discourse.*

No fifth structural locus remains under the fixed downstream substrate.

Proof. Take any interpretation package $\mathfrak{I}$. If it is not discourse-committing, it is discourse- deflationary and we are in class (4).

Assume instead that $\mathfrak{I}$ is discourse-committing. By Lemma 8.6, it must stabilize admissible reference assignments strongly enough to avoid quotient-relevant divergence. If the stabilizing work is done by changing the physical dynamics so that only one continuation remains eligible, the package realizes dynamic selection. If the stabilizing work is done by additional ontological structure not fixed by the bare dynamics, the package realizes ontic privileging. If the physical theory remains non-selective and the stabilization is achieved only by restricting admissible representatives to one quotient class, the package realizes semantic restriction. By Theorem 8.12, no fifth same-scope selector route exists. Hence the listed classes are exhaustive and mutually exclusive. □

8.3 The comparative interpretation matrix

Class	Selector locus	What is fixed	Representative strategies
Dynamic selection	Physical dynamics	One continuation by law	Collapse dynamics, objective reduction
Ontic privileging	Added ontology	One continuation by extra physical fact	Hidden-variable, actual-history, privileged trajectory
Semantic restriction	Admissible reference assignments	One standing-equivalence class of admissible assignments	Everettian semantic-fixation programs, typed successor restrictions
Discourse deflation	Nowhere inside the theory	The target discourse is refused	Instrumentalism, QBism, other deflationary responses

Table 2: Comparative interpretation matrix. The classification question is not which interpretation is metaphysically best, but where the work of fixing singular reference is done if it is done at all.

8.4 Where the familiar interpretations land

Proposition 8.14 (Collapse strategies instantiate dynamic selection)**.** *Collapse strategies such as GRW instantiate class (1) of Theorem 8.13: they place the work of singular-reference fixation in the physical dynamics itself.*

Proof. A collapse theory changes the dynamics so that, for the relevant measurement episode, one outcome history is physically singled out rather than multiple stable histories remaining equally realized. That is exactly dynamic selection. □

Proposition 8.15 (Hidden-variable strategies instantiate ontic privileging)**.** *Hidden-variable or actual-history strategies instantiate class (2) of Theorem 8.13: they place the work of singular-reference fixation in added ontological structure rather than in the bare dynamics alone.*

Proof. Such strategies retain a multiplicity in the wavefunction-level description but add further structure — for example an actual configuration or privileged trajectory — that designates which history is the one relevant to diachronic discourse. That is ontic privileging. □

Proposition 8.16 (Bare Everett is not yet a completed discourse-committing strategy)**.** *The bare Everettian package is physically non-selective and therefore does not by itself realize class (1),*

class (2), or class (3). To underwrite the target discourse, it must either be completed by semantic restriction or move to discourse deflation.

Proof. Theorem 4.8 and Corollary 4.7 show that the bare Everettian package is physically non-selective. So it does not realize dynamic selection. By construction it adds no extra ontology that privileges one branch, so it does not realize ontic privileging. It realizes semantic restriction only if admissible reference assignments are restricted strongly enough to collapse into one standing-equivalence class. Bare Everett alone does not do that. Therefore it is not yet a completed discourse-committing strategy. To underwrite the target discourse, it must either be completed by semantic restriction or retreat to discourse deflation. □

Corollary 8.17 (Bare Everettian exclusion of the first three loci)**.** *If one retains the bare Everettian package without adding new dynamics, without adding ontological selection structure, and without imposing an admissible same-scope restriction on reference assignments, then the package realizes none of the first three loci of Theorem 8.12. Relative to the target explanatory practice, only discourse deflation remains.*

Proof. Theorem 4.8 excludes dynamic selection in the bare Everettian package. Proposition 8.16 excludes ontic privileging and shows that the bare package does not by itself collapse admissible assignments into a single standing-equivalence class. Therefore, absent additional same-scope structure, the bare package realizes none of the first three loci of Theorem 8.12. If the target explanatory practice is nonetheless refused, the package moves to discourse deflation. □

Proposition 8.18 (Everettian semantic-fixation strategies instantiate semantic restriction)**.** *Any Everettian strategy that keeps the physical theory non-selective while restricting admissible diachronic reference strongly enough for records to function as evidence falls under class (3) of Theorem 8.13.*

Proof. Such a strategy leaves the underlying physical package non-selective, so it is neither dynamic selection nor ontic privileging. If it still underwrites theory-level confirmation, memory, anticipation, and agency, then by Lemma 8.6 it must stabilize admissible reference assignments. Since the work is not being done by a physical selector, it is being done by restricting admissible assignments themselves. That is exactly semantic restriction. □

Proposition 8.19 (QBist and instrumentalist strategies instantiate discourse deflation)**.** *QBist and instrumentalist strategies [14] fall under class (4) of Theorem 8.13 insofar as they do not take quantum theory to underwrite the target class of truth-apt diachronic measurement discourse.*

Proof. On QBist and related instrumentalist views, quantum states and outcomes are not taken to describe an objective diachronic measurement history in the sense needed by the standing conditional. Instead they organize expectations, betting commitments, or instrumental predictions. The package therefore does not aim to secure singular reference for the target discourse through the theory itself. That is not a failed stabilization strategy; it is a refusal of the target explanatory burden. Hence these views instantiate discourse deflation. □

Remark 8.20 (Relational and perspectival views)**.** Some relational or perspectival views may not advertise themselves under any of the four standard labels [15]. The present taxonomy still applies. If such a view is discourse-committing and physically non-selective, it must either supply semantic restriction or add physical selection. If it instead treats the relevant diachronic discourse

as non-fundamental or merely perspective-bound in a way that refuses theory-level truth-aptness, it moves toward discourse deflation. The exact placement of particular relational programs is therefore an application question, not a challenge to the taxonomy.

9 Objections as denial conditions

At this point the relevant objections are no longer free-floating. Each survives only by denying a labeled definition, lemma, or theorem. Table 3 states the denial burden in its strict form before the representative objections are spelled out in brief.

Table 3: Strict denial table for the main hostile-referee objections.

Objection	To maintain it, one must deny	Main location
The framework is optional	Theorem 2.8 or the standing conditional fixed in Section 2.	Sec. 2
You chose the quotient	Theorem 3.14.	Sec. 3
Alternative same-scope equivalence relation remains available	Theorem 3.14 together with Corollaries 3.15 and 3.16.	Sec. 3
Branch-local determinacy is enough	Theorem 4.11.	Sec. 4
Everett still contains an implicit selector	Theorem 8.12 or Corollary 8.17.	Sec. 8
Confirmation can be relativized or weakened	Theorems 5.7 and 3.22.	Sec. 5
Agency, memory, and anticipation are separate issues	Theorems 7.5, 7.12, and 7.13.	Sec. 7
There may be a fifth interpretive family	Theorems 8.8 and 8.12.	Sec. 8
An alternative selector outside the taxonomy remains available	Theorems 8.8 and 8.12.	Sec. 8
QBism or instrumentalism refutes the theorem ladder	Proposition 9.8.	Sec. 9

No row in Table 3 is rhetorical. Each identifies the smallest formal denial required to keep the

corresponding objection alive on the fixed scope. The brief subsections that follow spell out several representative rows.

9.1 Objection 1: branch-relative determinacy is enough

Proposition 9.1 (Branch-relative determinacy does not suffice for the target discourse)**.** *To maintain that branch-relative determinacy is sufficient, one must deny Lemma 4.11 or restrict the target discourse so that it no longer asks for theory-level diachronic reference.*

Proof. Lemma 4.11 proves that branch-relative determinacy and diachronic referential determinacy are distinct. The target explanatory practice of Definition 2.12 concerns the latter, not merely the former. So branch-relative determinacy is enough only if one either denies that distinction or weakens the target discourse. □

9.2 Objection 2: this is only self-location in different language

Proposition 9.2 (Self-location presupposes typed successor reference)**.** *To maintain that the problem is only self-location, one must deny Theorem 7.10 or show that self-locating discourse does not presuppose typed successor and record reference.*

Proof. Theorem 7.10 shows that self-locating and decision-theoretic discourse presuppose typed successor reference. So the self-location response is not a prior solution unless that dependence is denied. □

9.3 Objection 3: decision theory solves the problem

Proposition 9.3 (Decision theory presupposes rather than creates reference fixation)**.** *To maintain that decision theory solves the reference problem, one must deny Theorem 7.10 or show that a decision rule can itself create the referential lineage to which it is applied.*

Proof. A decision rule distributes weights or preferences only over already typed future alternatives. Theorem 7.10 formalizes exactly that dependence. So a solution by decision theory alone requires denial of that theorem. □

9.4 Objection 4: reference indeterminacy need not break science

Theorem 9.4 (Unresolved admissible divergence destabilizes theory-level science)**.** *Under the standing conditional of Section 2, unresolved admissible divergence in reference assignments for measurement records makes theory-level confirmation, and with it the memory/anticipation/agency package built around those records, assignment-relative rather than theory-relative.*

Proof. Corollary 5.8 proves that quotient-relevant divergence among admissible assignments makes comparative confirmation assignment-relative. Theorem 7.14 proves that memory, anticipation, and agency inherit the same lineage dependence. So unresolved admissible divergence does not leave the target scientific practice intact. It makes the central uses of measurement records assignment-relative. □

9.5 Objection 5: semantic infrastructure is unnecessary

Proposition 9.5 (Semantic infrastructure is required by the standing conditional)**.** *Within the target class fixed by the standing conditional, a semantic infrastructure constraint is not an optional embellishment. To deny it, one must deny Theorem 5.9 or reject the target explanatory practice.*

Proof. Theorem 5.9 states that a discourse-committing, physically non-selective theory must restrict admissible reference assignments strongly enough for fixation commutation to hold. A semantic infrastructure constraint is exactly such a restriction in the non-dynamic, non-ontic case. So denying the need for semantic infrastructure requires denying the theorem or exiting the target discourse. □

9.6 Objection 6: the comparison with collapse and hidden-variable views is unfair

Proposition 9.6 (The comparative matrix is classificatory rather than adversarial)**.** *The selector taxonomy does not privilege one interpretation family by rhetoric. To reject the comparative matrix, one must identify a fifth same-scope locus of singular-reference fixing not covered by Theorem 8.13.*

Proof. Theorem 8.13 classifies every package by the same structural question: where is singular reference fixed, if it is fixed at all? Propositions 8.14, 8.15, 8.18, and 8.19 merely place familiar interpretation families into that common matrix. So the classification is unfair only if a fifth same-scope locus exists. The theorem denies exactly that. □

9.7 Objection 7: the paper confuses semantics with physics

Proposition 9.7 (The result is conditional on explanatory use)**.** *The paper does not replace physical analysis with semantic legislation. To maintain the opposite, one must deny the conditional fixed in Section 2: namely, that the theory is being used to underwrite truth-apt diachronic discourse through measurement records.*

Proof. Definitions and theorems in Sections 2–7 all take a physical package as given and then ask what follows if that package is assigned a particular explanatory burden. If that burden is removed, the conditional result does not apply. That is not a confusion between semantics and physics; it is the explicit scope condition of the paper. □

9.8 Objection 8: instrumentalism or QBism dissolves the problem

Proposition 9.8 (Discourse deflation exits the standing conditional)**.** *Instrumentalist or QBist strategies do not refute the theorem ladder. They exit it by refusing the standing conditional under which the paper runs.*

Proof. Proposition 8.19 places QBist and related instrumentalist strategies in class (4) of Theorem 8.13: discourse deflation. Such strategies do not claim that the theory underwrites the target class of truth-apt diachronic measurement discourse. Because the standing conditional is refused, the need for reference stabilization inside the theory is likewise refused. That makes the strategy a coherent exit from the target problem, not a counterexample to the conditional theorem ladder. □

9.9 What remains open

Four things remain open and are consistent with the theorem ladder.

1. The paper does not prove that a successful Everettian semantic-fixation strategy exists. It proves only that some such restriction is required if the target discourse is to remain theory-relative.
2. The paper does not rank the interpretation families by independent metaphysical virtues. The taxonomy is structural.
3. The paper does not deny that local branch-internal discourse may remain fully determinate once a branch is held fixed.
4. The paper does not deny the coherence of discourse-deflationary responses. It says only that they are exits from the target explanatory burden rather than satisfactions of it.

10 Scope and non-claims

10.1 What the manuscript has established

On the fixed downstream substrate and within the standing conditional of Section 2, the manuscript has established the following.

1. a gate-governed fixed-domain definition of admissible reference assignment;
2. a measurement-reference quotient theorem identifying the unique standing-bearing content space for admissible reference assignments;
3. a corollary that confirmational-role divergence is quotient-relevant rather than semantic skin;
4. an Everettian non-selection theorem for the bare unitary, entangling, decohering package;
5. a confirmation criterion tying theory-level confirmation to invariance on the fixed measurement-reference quotient;
6. a fixation-commutation necessity theorem for theory-level confirmation;
7. a toy witness showing hypothesis-order reversal under divergent admissible assignments;
8. a lineage theorem showing that memory, anticipation, deliberation, and confirmation share one identity-preserving referential dependence; and
9. a structural exhaustion theorem locating every interpretation package at exactly one locus of singular-reference fixing, if it fixes singular reference at all.

10.2 What the manuscript does not claim

The manuscript does *not* claim any of the following.

1. It does not derive the Born rule, the Hilbert-space formalism, or any specific quantum probability law.

2. It does not prove that Everettian physics is false. It proves only that the bare Everettian package is physically non-selective with respect to the target discourse.
3. It does not settle every metaphysical question in quantum foundations. The taxonomy is structural, not a total ranking of interpretations by all virtues.
4. It does not deny branch-local determinacy. It denies only that branch-local determinacy by itself fixes diachronic reference on the target scope.
5. It does not claim that every scientific use of quantum theory collapses if reference is underfixed. Its stronger conclusion is restricted to the discourse named in the standing conditional: memory, anticipation, deliberation, and theory-level confirmation through measurement records.
6. It does not treat semantic infrastructure constraints as extra dynamical laws or hidden variables. They remain restrictions on admissible reference assignments imposed by the explanatory work the theory is asked to do.
7. It does not deny the coherence of discourse-deflationary responses. It says only that they refuse the target explanatory burden rather than satisfy it.

10.3 Role of these limits

These limits keep the target of the paper exact. The manuscript is a fixed-domain quotient-and-uniqueness paper about measurement-record discourse in physically non-selective theories. Its aim is not to settle every question in quantum foundations, but to isolate the referential condition under which measurement records can function as theory-level evidence at all.

11 Conclusion

The main result is conditional but exact. The paper does not claim that quantum mechanics requires new dynamics, nor that Everettian physics is false. It claims something narrower and more basic. If a physically non-selective theory is used to underwrite truth-apt diachronic discourse in which measurement records function as evidential anchors, then admissible reference assignments for those records must collapse into a single standing-bearing class on the fixed scope. If they do not, then confirmation, memory, anticipation, and deliberation become assignment-relative rather than theory-relative.

The theorem ladder reaches that conclusion through four steps. First, admissible reference assignments are typed in fixed-domain form and their standing-bearing content is shown to factor through a unique measurement-reference quotient. Second, the bare Everettian package is shown to be physically non-selective relative to the target discourse. Third, theory-level confirmation is shown to require fixation commutation under admissible redescription; quotient-relevant divergence therefore makes confirmation undefined on the fixed scope rather than merely weaker. Fourth, memory, anticipation, deliberation, and confirmation are shown to share one identity-preserving referential lineage through the relevant measurement records. The interpretation taxonomy is then a downstream exhaustion result about where that lineage can be fixed, if it is fixed at all.

The resulting diagnosis differs from the standard presentation of the measurement problem. On the target discourse, the first failure is not yet that a physically non-selective theory does not

produce one outcome. The first failure is that the theory may leave underfixed the referents of the very records through which outcomes are reported, remembered, anticipated, and used as evidence. In that precise sense, the measurement problem is a problem of reference to measurement records before it is a problem of outcome production.

A Appendix: formal fixed-scope quotient presentation

This appendix gives a compact formal presentation of the quotient machinery used in the main body. Its job is narrow: to state in one place the fixed scope, the admissible assignment space, the standing equivalence relation, the quotient universal property, and the reason confirmational-role divergence is quotient-relevant rather than semantic skin.

A.1 Fixed scope and admissible assignment space

Fix a physically non-selective measurement setting on the downstream substrate of Definition 2.1. Let ARef be the set of admissible reference assignments from Definition 3.3. For each $\rho \in \mathsf{ARef}$ and for each claim $d \in \mathcal{D}$, let $\mathsf{Val}_\rho(d)$ denote the standing-bearing discourse value of d on the fixed scope.

Definition A.1 (Target-discourse profile map)**.** The *target-discourse profile map* is

$$\mathsf{Prof} : \mathsf{ARef} \to \prod_{d \in \mathcal{D}} \mathcal{V}_d, \qquad \mathsf{Prof}(\rho) := \big(\mathsf{Val}_\rho(d)\big)_{d \in \mathcal{D}},$$

where $\mathcal{V}_d$ is the codomain of the standing-bearing value attached to d.

Definition A.2 (Standing equivalence on admissible assignments)**.** For $\rho, \sigma \in \mathsf{ARef}$, define

$$\rho \approx_{\mathrm{MR}} \sigma \iff \mathsf{Prof}(\rho) = \mathsf{Prof}(\sigma).$$

The quotient of admissible assignments by this relation is written $\mathsf{ARef}/\approx_{\mathrm{MR}}$.

A.2 Presentations and quotient uniqueness

Definition A.3 (Standing presentation)**.** A *standing presentation* for admissible measurement-reference content is a pair (X, r) such that $r : \mathsf{ARef} \twoheadrightarrow X$ and every standing-bearing value on the fixed scope factors through r.

Lemma A.4 (Automatic completeness)**.** *Every standing presentation is complete: if $r(\rho) = r(\sigma)$, then $\rho \approx_{\mathrm{MR}} \sigma$.*

Proof. If $r(\rho) = r(\sigma)$, then every standing-bearing value factors through the same point of X. Hence the values of every claim in $\mathcal{D}$ agree under ρ and σ. So $\mathsf{Prof}(\rho) = \mathsf{Prof}(\sigma)$ and therefore $\rho \approx_{\mathrm{MR}} \sigma$. □

Lemma A.5 (Automatic soundness on the fixed downstream substrate)**.** *Every standing presentation is sound on the fixed downstream substrate: if $\rho \approx_{\mathrm{MR}} \sigma$, then $r(\rho) = r(\sigma)$.*

Proof. Suppose $\rho \approx_{\mathrm{MR}} \sigma$ but $r(\rho) \neq r(\sigma)$. Then the presentation distinguishes two same-scope contents that agree on every standing-bearing value in the target discourse. Within the fixed downstream substrate, any same-scope admissibility-bearing distinction must survive either into standing classification or into licensed continuation. Since no such difference survives by hypothesis, the distinction is not standing-bearing. Treating it as such therefore violates the fixed-domain uniqueness package. Hence $r(\rho) = r(\sigma)$. □

Lemma A.6 (Quotient universal property)**.** *Let $q : \mathsf{ARef} \to \mathsf{ARef}/\approx_{\mathrm{MR}}$ be the quotient map. If $F : \mathsf{ARef} \to Y$ is constant on $\approx_{\mathrm{MR}}$-classes, then there exists a unique map $\widetilde{F} : \mathsf{ARef}/\approx_{\mathrm{MR}} \to Y$ such that $F = \widetilde{F} \circ q$.*

Proof. Define $\widetilde{F}([\rho]) := F(\rho)$. This is well-defined because F is constant on the $\approx_{\mathrm{MR}}$-classes. Uniqueness follows from surjectivity of q. □

Theorem A.7 (Quotient uniqueness)**.** *Let (X, r) be any standing presentation for admissible measurement-reference content. Then there is a unique bijection*

$$\phi : \mathsf{ARef}/\approx_{\mathrm{MR}} \longrightarrow X, \qquad \phi([\rho]) = r(\rho).$$

Hence $\mathsf{ARef}/\approx_{\mathrm{MR}}$ is the unique same-scope content space for standing-bearing measurement-reference assignments, up to unique isomorphism.

Proof. Completeness is Lemma A.4 and soundness is Lemma A.5. The proof of bijectivity is then identical to the proof of Theorem 3.12: soundness gives well-definedness, completeness gives injectivity, surjectivity comes from surjectivity of r, and uniqueness is the quotient universal property of Lemma A.6. □

Theorem A.8 (Global equivalence collapse on the fixed scope)**.** *Let $\approx$ be any same-scope equivalence relation on ARef through which every standing-bearing value factors. Then for all $\rho, \sigma \in \mathsf{ARef}$,*

$$\rho \approx \sigma \iff \rho \approx_{\mathrm{MR}} \sigma.$$

So there is no finer or coarser same-scope equivalence relation that preserves standing-bearing measurement-reference content.

Proof. If $\approx$ were coarser than $\approx_{\mathrm{MR}}$, it would identify assignments with different profile values, so the profile map would fail to factor through $\mathsf{ARef}/\approx$. If it were finer than $\approx_{\mathrm{MR}}$, it would distinguish assignments that agree on every standing-bearing value, thereby introducing a second same-scope closure-relevant equivalence relation. Both are impossible on the fixed downstream substrate. Hence the relations coincide. □

A.3 Why confirmational divergence is not semantic skin

Corollary A.9 (Confirmational-role divergence is quotient-relevant)**.** *If two admissible assignments $\rho, \sigma \in \mathsf{ARef}$ disagree on the comparative confirmational role of any evidential claim, then they belong to distinct classes in $\mathsf{ARef}/\approx_{\mathrm{MR}}$.*

Proof. Comparative confirmational role is one coordinate of the target-discourse profile. So a difference in that coordinate implies $\mathsf{Prof}(\rho) \neq \mathsf{Prof}(\sigma)$ and hence $\rho \not\approx_{\mathrm{MR}} \sigma$. □

Remark A.10. This is the formal point that blocks the easiest philosophical dismissal of the paper. The manuscript is not merely claiming that two semantic descriptions feel different. It proves that if they differ in confirmational role, then the difference survives the unique same-scope quotient of admissible measurement-reference content. That is standing-bearing divergence, not semantic skin.

B Appendix: companion structural package and exact imports

This appendix states exactly which companion results are used as fixed-domain background and how they are used. The point is transparency, not retrospective promotion. The present manuscript is conditional on these results; they are not hidden premises.

B.1 Imported fixed-domain package

The main body uses the following companion results only at the level stated below.

1. **Kernel closure and non-derivability.** Admissibility, standing, reference, and irreversibility are forced for the target class of non-degenerate reasoning, and there is no faithful same-scope lower generator.
2. **AMetric boundary and gate uniqueness.** There is no admissibility-relevant parameterization, selector, ranking, or same-scope competing gate on the fixed scope.
3. **Standing conservation.** There are no same-scope repairs, generators, or primitive carriers, and no transfer without fresh target gating.
4. **Unique admissible interior.** The admissible interior is unique up to lawful equivalence.
5. **Trajectory closure.** Same-scope standing failure is irrecoverable under admissible continuation.
6. **No second invariant / no fourth role.** Any second same-scope invariant collapses to bookkeeping or scope change, and any admissibility-preserving role classification factors through anchor/tensor/skin up to role isomorphism.

The relevant companion manuscripts are [16–23].

B.2 How those imports are used here

The imported package is used in exactly four places.

1. In Section 2, it fixes the downstream substrate and blocks the "your framework is optional" objection.
2. In Section 3, it forces soundness of measurement-reference presentations and therefore yields quotient uniqueness on the fixed scope.
3. In Section 5, it turns confirmational divergence into a standing-bearing difference rather than a mere semantic redescription.
4. In Section 8, it closes the no-fifth-locus theorem by ruling out a second same-scope admissibility classifier, a second invariant, or a fourth admissibility-bearing role.

No proof in the main body uses the companion papers for anything stronger than these fixed-domain constraints.

B.3 What the present paper adds beyond the companion package

The companion papers do not by themselves solve the measurement problem. What the present paper adds is a fixed-domain quotient-and-uniqueness analysis of measurement-record discourse in physically non-selective theories. Its distinctive results are:

1. the explicit admissible assignment space for diachronic measurement-record discourse;
2. the measurement-reference quotient theorem;
3. the Everettian non-selection theorem in this measurement-specific setting;
4. the fixation-commutation criterion for theory-level confirmation; and
5. the common-lineage theorem for confirmation, memory, anticipation, and deliberation.

So the relation between the present paper and the companion corpus is asymmetric. The companion results fix the downstream substrate; the present paper applies that substrate to measurement-record reference.

C Appendix: audit materials

This appendix collects three audit aids: a dependency map, a denial-condition matrix, and a compact reading path. It adds no premises to the theorem ladder.

C.1 Dependency map for the theorem ladder

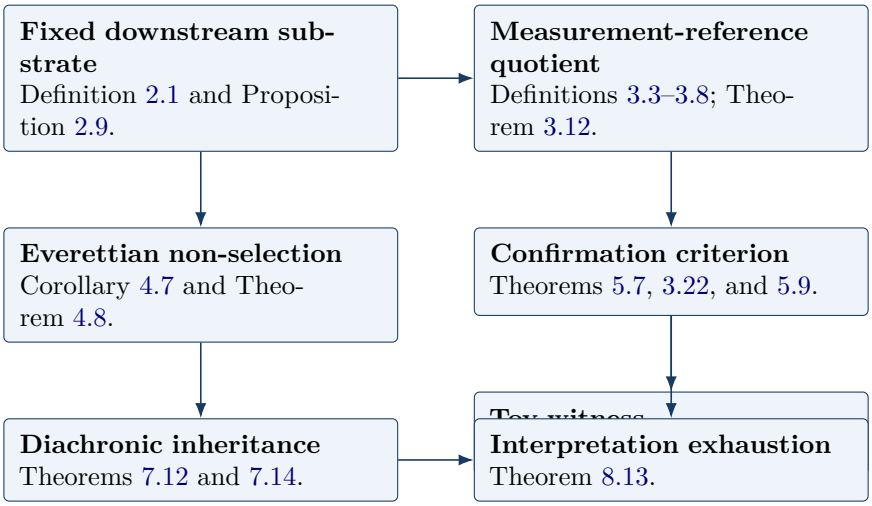

Figure 2: Dependency map for the hardened theorem ladder. The quotient theorem and the Everettian non-selection theorem are the two main inputs to the confirmation and lineage theorems; the interpretation taxonomy is downstream of both.

C.2 Denial-condition matrix

Table 4: Denial-condition matrix for standard objections.

Objection	To maintain it, one must deny	Where to look
The framework is optional	The compressed substrate-necessity bridge or the standing conditional itself	Sec. 2
Different admissible assignments are just semantic packagings	The equivalence-collapse theorem or Corollary 3.18	Sec. 3; App. A
Branch-local determinacy is enough	Lemma 4.11	Sec. 4
Decision theory solves the problem	Theorem 7.10	Sec. 7
Confirmation can be relativized without loss	Theorems 5.7 and 3.22	Sec. 5
Memory, anticipation, and agency are separate issues	Theorems 7.12 and 7.13	Sec. 7
There may be a fifth interpretive family	Theorem 8.12	Sec. 8
Instrumentalism or QBism refutes the theorem ladder	Proposition 9.8	Sec. 9

C.3 Compact reading path

A short but faithful route through the manuscript is:

1. Section 2: fixed downstream substrate, target discourse, and standing conditional.
2. Section 3: Definitions 3.3–3.8, Theorem 3.12, and Corollary 3.18.
3. Section 4: Corollary 4.7 and Theorem 4.8.
4. Section 5: Theorems 5.7, 3.22, and 5.9.
5. Section 6: Theorem 6.3.
6. Section 7: Theorems 7.12 and 7.14.
7. Section 8: Theorem 8.13.
8. Section 9: only the objection subsections still thought live after the first seven steps.

C.4 Role of the appendix

This appendix is procedural. The dependency map, denial-condition matrix, and reading path expose the architecture of the argument without adding premises to it. Table 3 in Section 9 is the main-text compression of the same hostile-referee logic.

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